COOLING TOWER MANUAL

Chapter 6

WATER TREATMENT

COOLING TOWER INSTITUTE
P.O. Box 73383 ● Houston, Texas 77273
(281) 583-4087 ● FAX (281) 537-1721
http://www.cti.org ● email: vmanser @cti.org

FOREWORD

This Cooling Technology Institute (CTI) publication is published as an aid to cooling tower purchasers and designers. It may be used by anyone desiring to do so, and efforts have been made by CTI to assure the accuracy and reliability of the data contained herein. However, CTI makes no warranty of fitness for particular purpose or merchantability nor any other warranty expressed, implied or statutory. In no event shall CTI be liable or responsible for Incidental, Consequential or Commercial losses or damage of any kind resulting from this publication's use; or violation of any federal, state, or municipal regulation with which this publication may conflict or for the infringement of any patent resulting from the use of this publication.

Nothing contained herein is to be construed as granting any right for the manufacture, sale or use in connection with any method, apparatus, or product covered by letters patent, nor as insuring anyone against liability for infringement of letters patent.

This guideline document summarizes the best current state of knowledge regarding the specific subject. This document represents a consensus of those individual members who have reviewed this document, its scope and provisions. It is intended to aid all users or potential users of cooling towers.

Approved by the CTI Executive Board.

This document has been reviewed and approved as part of CTI's Five Year Review Cycle. This document is again subject to review in 2010.

Chapter 6

Table of Content

Chapter 6
Water Treatment

Section I.
INTRODUCTION

Water is an effective heat transfer agent. It is also an excellent solvent and therefore contains amounts of almost every substance with which it has been in contact since it fell from the sky, yesterday or 1,000 years ago. In addition, water facilitates the dissolution of the components of mechanical systems that it contacts. Water is also a desirable biological media. Water will shelter and transport a wide range of microbes, the nutrients needed for growth. Effectively utilizing water in recirculating cooling water systems without experiencing unacceptable rates of equipment degradation and performance deterioration requires understanding and control of system water chemistry. This also involves proper equipment design to minimize the potential for water related problems. In addition, water source selection, water pretreatment before use and internal water treatment are necessary.

The water chemistry, treatments and operational control methods discussed in this chapter apply specifically to open recirculating cooling water systems containing either cooling towers or evaporative condensers. Heat rejection in open recirculating systems is accomplished primarily by evaporation of water in the cooling tower. The air-water contact in the cooling tower affects, both directly and indirectly, the system water chemistry. Direct contact of water with air results in near saturation with dissolved oxygen and carbon dioxide in the air. Airborne dust is scrubbed from the air, increasing the suspended solids concentration of the cooling water, and the water is continuously inoculated with airborne microorganisms. Evaporation of water in the cooling tower causes the dissolved and suspended matter in the makeup water to be concentrated in the recirculating water. These processes all influence the corrosion, scaling, deposition and microbiological fouling potential of the system.

The objective of this chapter is to provide a general overview of open recirculating cooling water operations, problems and chemical treatment, as a technical resource and reference aid for process engineers and management personnel with no previous experience in cooling water systems. After studying this chapter, workers who need more detailed technical information will find many good textbooks and reference papers available. At the other end of the spectrum, operating personnel should refer to specific process manuals for detailed operating information.

Section II.
SOURCES OF WATER

Most of the world's available water supply comes from moisture that has evaporated from oceans, seas, rivers, lakes, land masses, etc. and then precipitated from the atmosphere. "Meteoric water" is used to describe this basic source of supply. Water may fall in the form of rain, snow, sleet or hail. As water precipitates from the atmosphere, it comes in contact with gases in the atmosphere as well as particles of dust, industrial smoke and volcanic dust. Water scrubs the gases and solids from the atmosphere. Several studies on meteoric water samples taken in the United States show significant amounts of dissolved minerals and a pH range of 3.5 to 7.1. Rain water is far from being "pure".

Surface water and ground water comprise the most important sources of fresh water. Some of the water that falls to the earth's surface soaks into the ground. This is ground water. Some water collects in ponds and lakes or runs off into streams and rivers. This is surface water.

Surface water carries finely divided particles such as silt, sand, etc and organic matter. The character of the area influences the nature and quality of the impurities found in the surface waters of a given geographic area.

Ground water is the portion of water that percolates through the earth and collects underground. This is where well and spring waters originate.

The composition of ground water is influenced by geographical variations in the aquifer from which the supply is drawn. Most underground supplies differ from surface supplies in the amount of suspended solids and dissolved solids. Generally, underground waters do not contain particulate matter or dissolved organic compounds, but may contain high levels of dissolved gas (CO_2) and minerals such as iron and manganese. The carbon dioxide content of ground water is much higher than that of surface water, due mostly to the decay of organic matter in the surface soil.

Other sources of water besides ground water and surface water are also important. These include reclaimed water and sea water. Industry attempts to recycle as much water as is feasible for economic, sociological and environmental reasons.

In many areas the availability of fresh water is limited. It is in industries' best interest to conserve available supplies by recirculating cooling water over cooling towers and through cooling systems. Many industrial plants are utilizing effluent streams from sewage plants for the makeup water. This is a good example of water reuse and conservation.

Sea water is also used for cooling applications. Industrial plants that are located on coastal waters and that have large cooling requirements can use sea water to their advantage because of the abundance of supply and relatively cool temperatures.

A careful analysis of the water supply is necessary so that appropriate purification and water treatment techniques can be used to produce water which is acceptable for industrial and public use. Table 1 lists some of the common impurities in natural water supplies and the problems that these substances can cause. Also included are the types of water treatments that are applied to alleviate the problems. This list is by no means a complete summary of all the analytical tests necessary for a rigorous examination of any given specific problem nor the exact treatment required to solve that problem.

.

TABLE I

MAJOR IMPURITIES, THEIR EFFECTS AND TREATMENT

(COURTESY BETZ LABS INC.)

Constituent	Problem	Treatment
Turbidity (Suspended Solids)	Imparts undesirable appearance to water. Deposits in water lines, process equipment, boilers, etc. interferes with most process uses.	Coagulation, settling and and filtration.
Color (Dissolved Organics)	Hinders precipitation methods such as iron removal, hot phosphate softening. Can stain product in process use.	Coagulation and filtration. Chlorination. Adsorption by activated carbon.
Hardness	Chief source of scale in heat exchange equipment, boilers, pipe lines, etc. Forms cruds with soap, interferes with dying, etc.	Softening. Distillation. Surface active agents. Internal cooling water treatment. Hydrogen zeolite softening. Demineralization.
Alkalinity	Calcium carbonate scale. High pH.	Lime and lime-soda softening. Acid treatment. Dealkalization by ion exchange.
Free Mineral Acidity	Corrosion.	Neutralization with alkalies. Weak base anion exchange
Carbon Dioxide	Corrosion in water lines.	Aeration. Deaeration. Neutralization with alkalies.
Sulfate	Adds to solids content of water. Combines with calcium to form calcium sulfate scale.	Demineralization. Distillation. Deposit control agents.

TABLE I (Continued)

MAJOR IMPURITIES, THEIR EFFECTS AND TREATMENT

(COURTESY BETZ LABS INC.)

Constituent	Problem	Treatment
Chloride	Adds to solids content and increases corrosivity of water.	Demineralization. Distillation.
Nitrate	Adds to solids content, but is not usually Distillation. significant industrially. High concentrations cause methemoglo-binemia in infants.	Demineralization.
Silica	Scale in cooling water systems, boilers, heat exchange equipment, etc. basic anion exchange resins in conjunction with demineralization. Distillation. Deposit control agents.	Hot process softening with magnesium salts. Adsorption by highly
Iron	Discolors water on precipitation. Source of deposits in water lines, heat exchangers, boilers, etc.	Aeration. Coagulation and filtration. Lime softening. Cation exchange. Contact filtration. Surface active agents for iron retention. Deposit control agents.
Manganese	Same as iron.	Same as iron.
Oil	Impedes heat exchange. Undesirable in most processes.	Baffle separators. Strainers. Coagulation and filtration. Diatomaceous earth filtration. Treatment with surfactants.
Oxygen	Corrosion of water lines, heat exchange equipment, return lines, etc.	Deaeration. Sodium sulfite. Corrosion inhibitors.
Hydrogen Sulfide	Cause of "rotten egg" odor.	Aeration. Chlorination. Highly basic anion exchange.

TABLE I (Continued)		
MAJOR IMPURITIES, THEIR EFFECT AND TREATMENT		
(COURTESY BETZ LABS INC.)		
Constituent	**Problem**	**Treatment**
Ammonia	Corrosion of copper and zinc alloys by formation of a complex soluble ion.	Cation exchange with hydrogen zeolite. Chlorination. Deaeration.
Dissolved Solids	"Dissolved solids" is a measure of the total amount of dissolved matter, determined by evaporation. High concentrations of dissolved solids are objectionable because of process interference. May also cause deposits in cooling systems.	Various softening processes, such as lime softening and cation exchange by hydrogen zeolite will reduce dissolved solids. Demineralization. Distillation.
Suspended Solids	"Suspended Solids" is a measure of undissolved matter, determined gravimetrically. Suspended solids plug lines, cause deposits in heat exchange equipment, boilers, etc.	Settling. Filtration, usually preceded by coagulation and settling. Polymer deposit control agents.
Total Solids	"Total Solids" is the sum of dissolved and suspended solids, determined by gravimetric analysis.	See "Dissolved Solids" and "Suspended Solids".

Section III.
EXTERNAL TREATMENT

External treatment means treatment or conditioning of a water supply to make the water more suitable for its intended application, before the water arrives at the point of use. External treatment is referred to by many different terms, including, for example, preliminary treatment, pretreatment, indirect treatment, primary treatment, etc.

External treatment includes both mechanical and chemical processes. Equipment may include hot or cold process lime-soda softeners, ion exchange systems, reverse osmosis units, aerators, filters, clarifiers, etc. These processes are used for reducing dissolved minerals and alkalinity, eliminating dissolved oxygen and reducing levels of suspended solids.

The selection of pretreatment processes for a given industrial water application requires joint technical and economic evaluations. For example, suspended solids can be removed by clarification and filtration, or dispersants can be used to prevent deposit formation in the cooling system. Hardness can be removed by lime or zeolite softening, or scale inhibitors and pH control can be used to control calcium carbonate scaling. The optimum choice in any system requires a good understanding of the technical problems and the relative capital and operating costs for the various options.

Aeration

Aeration is generally used to remove (strip) undesirable gases from water, aid in removing undesirable metals such as iron by oxidation, eliminate odors, tastes and colors and lower overall chemical costs of water treatment.

Aeration intimately mixes water and air. This mechanical process can be accomplished by forming thin films, drops or a spray of water. The main objective is to achieve equilibrium between the gases in the water and the gases present in the air.

In industrial cooling water treatment, one of the ways carbon dioxide is removed is via the cooling tower. A cooling tower will strip carbon dioxide from the recirculating water. In addition, other soluble gases such as hydrogen sulfide, which may be a contaminant in the cooling water, can also be "aerated off". However, such gases can be a source of noxious odors. The industrial cooling tower is similar in many design aspects to a water-fall aerator.

The concept of the cooling tower as a mechanical aerator is an important concept to remember. Various dissolved impurities such as iron and manganese, when aerated through the cooling tower, can pose a deposition problem, due to the formation of insoluble metal oxides.

Although the cooling tower can act as an aeration point, it also is responsible for saturating the water with dissolved oxygen. This, in turn, increases the corrosivity of the recirculating water. The removal of other corrosive gases such as hydrogen sulfide may not be complete unless aeration is combined with pH reduction or chlorination. In the case of the cooling tower, this is not always practical because such changes can adversely affect the performance of the cooling water chemical treatment program.

Clarification

Clarification is a general term for a group of combined mechanical and chemical processes used to remove suspended solids such as sand, silt and naturally occurring organics from water supplies. Suspended solids levels in water supplies often need to be reduced to make the water suitable for industrial purposes. Clarification includes three specific steps: coagulation, flocculation and sedimentation.

Coagulation is the process of destabilizating colloidal suspended solids by charge neutralization. Once this occurs, the particles no longer repel each other, and hence can come together to form larger, settleable solids. Moderate molecular weight cationic polyelectrolytes are used to neutralize the negative charges on most naturally occurring suspended matter.

Flocculation is the process of bringing together, or "bridging" the coagulated particles to form larger particles or "floc".

Sedimentation is the physical settling that occurs after the particles have been coagulated and flocculated. Settling during the sedimentation process can only remove relatively coarse suspended solids without prior coagulation, and/or flocculation.

Each water supply must be tested to determine the optimum clarification process needed for suspended solids removal. Some suspended solids will settle and provide adequate clarity in the water without chemical treatment. Other solids need coagulation or charge neutralization to form a settleable floc, while in other cases, a flocculation, or bridging step is required.

<u>Chemical Selection</u>

In-field jar testing is normally used to define the chemicals and dosages needed for a specific clarification process. Since the suspended solids content of a water supply can change seasonally and also on a daily or even hourly basis, it is important to run tests frequently enough to be sure that chemical feed rates are optimized. Automatic instruments that measure either the incoming water turbidity or the net negative charge on colloidal particles are often used to control chemical dosage rates in large clarification plants.

<u>Upflow Clarifiers</u>

Upflow clarifiers offer a compact and economical way to manage coagulation, flocculation and sedimentation processes. These clarifiers are usually circular in shape and made of steel or concrete. The water flows upwards, which is a key feature in maintaining high clarity of the outlet water. This is possible because these units maintain good sludge bed solids contact through internal sludge recirculation.

Filtration

Filtration is a mechanical process for removing suspended matter from water by collecting the solids on a porous medium. Filtration does not remove dissolved or colloidal solids. Most clarification or softening processes, in which coagulation and/or precipitation occur, process at least a portion of the water through filters.

Most rapid flow filters process the water flow downward. Both gravity and pressure filters typically operate in this manner. The filter bed usually consists of sand or anthracite or various mixed grades with a total depth of 15 inches to 30 inches.

Upflow filters, on the other hand, pass the water up through a single medium bed. The major disadvantage of upflow filters is the difficulty in backwashing the bed. Downflow washing of an upflow filter produces no bed expansion or scouring action. Even though upflow washing is effective, it can leave a portion of the coarse dirt trapped in the lower layers of the filter media. Another drawback is that excessive bed expansion during operation limits the maximum rate at which water can be filtered satisfactorily.

In-line Clarification

In-line clarification utilizes filtration for removal of makeup water turbidity. The major difference between in-line clarification and standard pressure filtration is that the entire bed removes floc particles instead of the top layer of sand. This is accomplished by pretreating the filter with cationic polyelectrolytes. These polymers adsorb on the filter bed and "catch" or neutralize colloidal particles as they pass by. Additional polymer is normally added to the water as it enters the filter.

Sidestream Filtration

Sidestream filtration simply means placing a filter in a bypass stream so that a portion of the total cooling water circulation rate (3% - 10%) is filtered. The advantages of sidestream filtration include lower capital and space requirements than in-line clarification. The complete cooling system can be "turned over" in one day. Sidestream filtration also has the advantage of being able to process the recirculating cooling water and remove debris which has been drawn in by the cooling tower, plus material which has precipitated in the bulk water. These filters are similar in general construction and design to normal gravity or pressure units. Since recirculating cooling water is normally used during backwashing, the backwash water is a source of cooling tower blowdown water.

Chemical Treatment in Filtration

Various chemicals are employed to improve filter performance. Surfactants are often added directly ahead of the filter to improve bed penetration, or during the backwash cycle to aid in scouring organics from the sand media. Polyelectrolytes, often called filter aids, are sometimes added during the backwash process to condition the media and allow for more efficient filtration. In addition, polyelectrolytes can also help settle the solids in a backwash tank, which can then be decanted so that the clear water can be returned to the cooling system.

Precipitation Softening

Chemicals can be added to makeup water to remove some specific unwanted soluble components by precipitation. Hot and cold lime softening are typical precipitation processes. These methods are suitable for softening high hardness and/or high alkalinity waters. Precipitation chemistry can also remove iron, manganese or suspended matter without the need for prefiltering the water.

The chemistry involved in lime or lime-soda softening processes is complex and beyond the scope of this manual. Briefly, lime (calcium hydroxide) is added to the water to react with bicarbonate alkalinity and convert it to carbonate. Since carbonate can react with twice as much calcium as bicarbonate, every part per million of calcium added as lime (expressed as carbonate) precipitates two parts per million as calcium carbonate, as long as sufficient bicarbonate alkalinity is present. If excess calcium is present, soda ash (sodium carbonate) can be added to precipitate the remaining calcium. This lime-soda softening calcium can be removed in this way down to the solubility of calcium carbonate (about 30 ppm at 70°F).

Well water supplies vary widely in calcium content. High calcium well waters may need precipitation softening or ion exchange softening to make them suitable for industrial cooling water use. Also, some reuse water supplies, such as wastewater and process water recycle, may require softening.

Typically, cold process lime or lime-soda softening is used to treat makeup water for cooling systems, prepare process water for the beverage industry and soften municipal supplies. Hot lime softening units are efficient for reducing alkalinity, hardness and silica before makeup water is fed to an ion exchange unit for boiler feedwater preparation.

Ion Exchange

Ion exchange resins are widely used to treat makeup water supplies that contain dissolved salts. These materials can exchange one ion for another, depending on relative concentrations in solution, hold it temporarily and then relinquish it to a regenerant solution. The ion exchange resin replaces undesirable ions in the water supply with other more acceptable ions. For example, sodium zeolite softeners can replace scale forming calcium and magnesium ions with soluble sodium ions. Zeolites contain strongly acidic cation sites in the sodium form.

Modern ion exchange materials consist of resin beads that form a hydrocarbon network to which ionizable functional groups are attached. For example, in cation resins, water diffuses into the beads and sodium (or hydrogen) cations are exchanged for the calcium and magnesium cations. The sodium (or hydrogen) ions diffuse into the water and out of the beads.

Commonly used industrial water treatment resins are classified in one of four basic types:

- Strongly basic anion.
- Weakly basic anion.
- Strongly acidic cation.
- Weakly acidic cation.

Strongly basic anion exchange resins obtain their functionality from quaternary ammonium exchange sites. These resins are generally used in demineralizer systems for boiler feedwater. Weakly basic anion resins derive their functionality from various amine groups. These resins adsorb such free mineral acids as hydrochloric and sulfuric acids, and are used in demineralizer systems in conjunction with strongly basic resins to reduce regenerant costs. These resins can also scavenge some organic matter from water. Anion resins can be regenerated with caustic (hydroxyl form) or salt (chloride form).

Strongly acidic cation resins derive their exchange activity from sulfonic acid functional groups. These resins will remove nearly all cations and function well at all pH ranges. These resins have found wide use for softening, dealkalization and demineralization of boiler feedwater. Weakly acidic cation exchange resins use carboxylic groups as the exchange sites. These resins remove hardness associated with alkalinity. These weakly acidic resins are used primarily for softening and dealkalinization and are frequently used with strongly acidic polishing resins. Cation resins can be regenerated with salt (sodium form) or a strong acid (hydrogen form).

Application of ion-exchange treated makeup water is limited to closed recirculating cooling water systems. Systems that require precise control of process temperatures, and those that contact very high skin and bulk water temperatures (greater than 200°F), such as jacket cooling in engine test stands, and steel mill continuous caster mold cooling systems, often utilize zeolite softened water. These systems do so economically through careful control of water losses. The use of ion exchange treated raw water for normal cooling tower makeup is generally unnecessary, and not economically feasible because of the large volume of water required for evaporative cooling.

Reverse Osmosis

Reverse osmosis is becoming widely accepted as a pretreatment step to improve the efficiency of ion exchange softening or demineralization when very hard or high dissolved solids makeup water can be used. Reverse osmosis makes use of special membranes permeable to water but not to dissolved ions. When these membranes are used to separate water streams with different dissolved solids contents, water will normally pass through the membrane from the low solid to the high solid side (osmotic flow). By applying pressure to the high solids side, this flow can be reversed so that water is forced to flow from high solids to low solids; hence the term "reverse osmosis".

Reverse osmosis can be used to prepare good quality deionized water, but it is most practical and economical as a roughing step to reduce the dissolved solids level in the makeup water prior to deionization. When used in this way, reverse osmosis may take the place of chemical precipitation (lime softening) processes. Reverse osmosis is rarely used as a pretreatment step for cooling water makeup, except when very low dissolved solids water is needed for closed systems.

Section IV.
FUNDAMENTALS OF OPEN RECIRCULATING COOLING WATER SYSTEMS

The primary function of all cooling water systems is to remove unwanted heat. This is true whether the heat is generated in a critical process reactor or in an executive office building. However, the fundamental principles of their operation and function are relatively consistent. Figure 1 illustrates the essential elements of an open recirculating cooling water system.

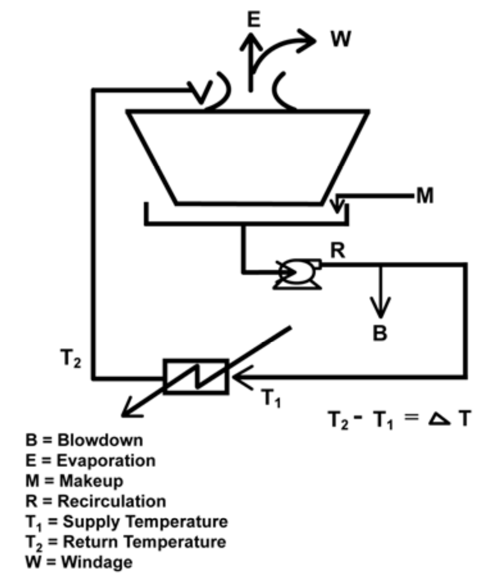

Figure 1
Open Recirculating Cooling Water System

In this design, cooling water is pumped from the cooling tower basin to a point of heat exchange or heat transfer, which increases the temperature of the cooling water from T_1 to T_2. The cooling water flows through the recirculating lines to the cooling tower deck, from which the water falls by gravity through the tower packing (or fill) to the tower basin. Heat is rejected to the atmosphere by means of evaporation (E) as the cooling water passes through the fill. Air passes upward by means of mechanical or natural draft. The rejection of heat in the recirculating water (R) by evaporation (E) restores the cooling water to temperature T_1 at which point the cooling cycle begins again. The cooling tower serves two purposes: heat rejection and water conservation.

In the operation of the cooling tower, water is continuously evaporated and must be replaced with fresh makeup water. This results in an increase in the concentration of dissolved solids in the recirculating water. The ratio of the total dissolved solids in the recirculating water to the total dissolved solids in the makeup water (TDS_r/TDS_m) is called cycles of concentration (C). As cycles of concentration (C) increase, some of the dissolved solids in the recirculating water approach the limit of their solubility in water. This is especially significant for dissolved minerals that form insulating scales, because these minerals frequently become less soluble as the temperature of the recirculating water increases (inverse or retrograde solubility). Therefore, they tend to deposit in areas of elevated temperature and lower water velocity, such as in heat exchange equipment. Insulating mineral scales are undesirable because they interfere with the efficient rejection of heat from the production process. This can result in reduced product throughput, poor product quality, and eventually lost production time for cleaning.

To prevent solids from continuing to concentrate in the cooling water and eventually precipitating and depositing on system components, a portion of the recirculating water is removed from the system and replaced with the less concentrated makeup water. The recirculating water removed from the system is referred to as blowdown (B). Cycles of concentration are controlled by the amount of blowdown.

In addition to blowdown, some water droplets containing concentrated dissolved solids are carried through the evaporation equipment in the cooling tower and lost to the atmosphere. This windage loss (W) may vary from one cooling tower to another, between 0.0005% and 0.2% of the recirculating rate (R). For most cooling tower calculations, windage (W) is included in the tower blowdown (B)*.

The water balance for the system in Figure 1 is:

$$M = E + B + W \text{ (or } M = E + B^*) \qquad (1)$$

Evaporation can be estimated for the system using the following expression:

$$E = .001 \times R \times T \times f \qquad (2)$$

This expression indicates that the evaporation rate (E) is approximately equal to 0.1% of the water circulated (R) over the tower for each 1°F of temperature drop (T_2 - T_1) times a factor (f).

The factor (f) represents the percent of cooling that is contributed by evaporation only. The total cooling effect when warm water is in contact with cooler air is not due to evaporation alone. Sensible heat transfer from the warm return water to the cooler air will accomplish a certain percentage of the cooling depending on the temperature of the ambient air. This convective cooling effect may range from 40% in winter to 10% in summer. Conversely, (f) will vary between 0.6 and 0.9, depending upon time of year.

To calculate the amount of blowdown water (B) required to maintain given cycles of concentration (C), use the equation:

$$B = E/(C-1) - W \text{ or} \qquad (3)$$

$$B^* = E/(C-1) \qquad (3a)$$

In practice, the quantity of makeup water required for the operation of a given cooling tower may be estimated with a fair degree of accuracy, using equations 1, 2 and 3. Cycles of concentration, (C), are frequently determined, not by total dissolved solids measurement, but by specific conductance or by any non-reactive chemical constituent which can be accurately tested in both the makeup and the recirculating water. Care should be taken that the measured constituent is not one that is added with the treatment chemicals. Chloride ion is often used but is not well-suited for this purpose when chlorination is practiced or other treatment products containing appreciable chloride are added to the system.

The concept of cycles of concentration is an important consideration in the operation of an open recirculating cooling system. The greater the number of cycles of concentration, the less water (and treatment) the system must lose to blowdown. In recent years some treatment programs have been able to achieve a zero blowdown condition in certain cooling systems, the cycles of concentration being limited by windage losses and leakage only. Figure No. 2 graphically illustrates the relationship between makeup water requirements and cycles of concentration, and Figures 3 and 4 show the chemical and water savings that can be realized by operating an open recirculating cooling water system at higher cycles of concentration.

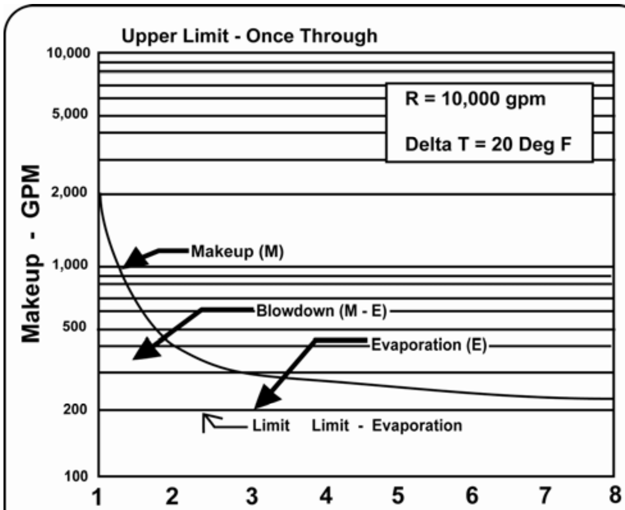

Figure 2
Makeup Flow As A Function Of Cycles of Concentration

In most open recirculating cooling water systems, the concentration of treatment in the cooling water required to inhibit scale and corrosion is relatively constant. Therefore, as cycles of concentration are increased to a maximum, the amount of treatment required to maintain a given concentration in the recirculating water is diminished proportionately. To determine the quantity of treatment needed, use the following equation:

---

* Includes windage (W), leaks, overflow losses, etc.

* B = Includes windage (W), leaks, overflow losses, etc.

lbs. treatment/unit time = (4)

$$\frac{M\ (\text{Gal/Unit Time}) \times \text{ppm (treatment)}}{C \times 120{,}000^{*}}$$

As cycles of concentration increase, the potential for corrosion, scaling and fouling generally increases. There is a practical limit beyond which cycles of concentration cannot be increased for most cooling systems. Potential problems and their control are discussed in a later Section V of this chapter.

The quantity of treatment required to slug dose a cooling system of V (Gal) is:

$$\text{lbs. treatment} = \frac{V\ (\text{Gal}) \times \text{ppm (treatment)}}{120{,}000} \quad (5)$$

It is frequently useful to determine the time required to reduce the level of a given species of treatment or of a contaminant in a cooling system:

$$t\ (\text{time}) = \frac{V(\text{Gal}) \times \ln (C_i/C_f^{**})}{B^{**}\ (\text{Gal/unit time})} \quad (6)$$

The half-life (t1/2) of a given species within a cooling system is a special case of equation 6.

$$t1/2\ (\text{time}) = .69\ V(\text{Gal})/B(\text{Gal/unit time}) \quad (7)$$

t1/2 is defined as the time required to reduce the concentration of a treatment chemical (or contaminant) by 50%, with no new treatment or contaminant in the makeup.

To determine the volume of a system, add 1/2 lb. table salt per 1,000 gallon of estimated volume. Measure the chloride level before addition and the peak chloride level after addition.

$$\text{Then } V\ (\text{gal}) = \frac{120{,}000 \times \text{lbs salt added}}{\text{ppm NaCl added}}$$

Example of the Use of Fundamental Calculations for Open Recirculating Cooling Systems

The following example illustrates the use of equations 1 through 7 in a typical open recirculating cooling water system.

An open recirculating cooling water system has a design recirculating rate of 12,000 gallons/minute and operates 24 hours per day year around.

The temperature drop measured across the cooling tower is 18°F, 108°F at the top of the tower and 90°F in the tower sump. The heat load on the system is constant. There are no water meters on the tower makeup line, but the blowdown line has been monitored and flow measured at 52 gallons/minute. The design specifications for the tower indicate windage losses at 0.025% of the recirculating rate.

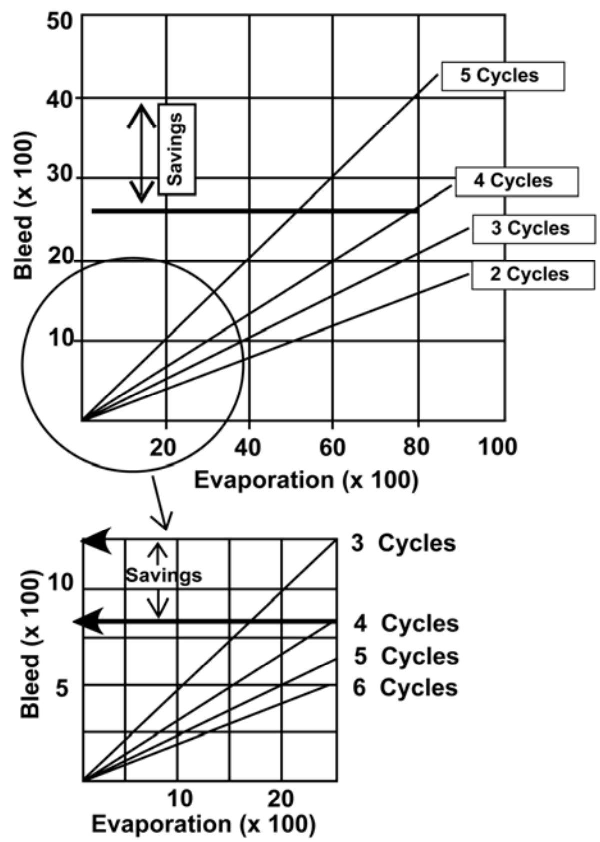

Figure 3
Water Savings From Increased Cycles

* 120,000 is merely a constant to convert Gal of blowdown to MM lbs. of blowdown; i.e., there are 120,000 Gal per million lbs. of water.

** C_i = initial concentration

C_f = final concentration

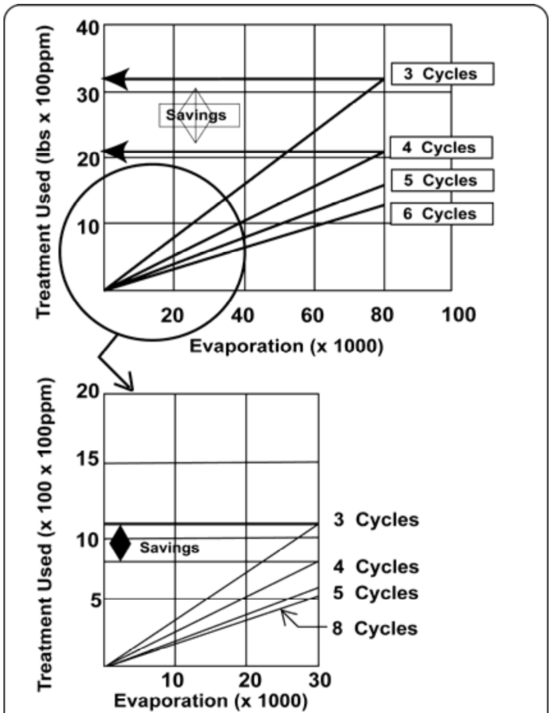

Figure 4
Chemical Savings From Increased Cycles

Use equation 2 to determine the evaporation rate (E):

$$E = .001 \times R \times T \times f$$

$$E = .001/°F \times 12{,}000 \text{ gal/min} \times 18°F \times 0.8 = 172.8 \text{ gal/min.}$$

f was estimated at 80% evaporative cooling

The specific conductance of the recirculating water measured 4 times that of the makeup water indicating that the system is operating at 4 cycles of concentration. This was confirmed by the silica concentrations as well.

The blowdown rate, then, is calculated using equation 3a. as follows:

$$B^* = E/(C-1)$$

$$B^* = 172.8 \text{ Gal/min}/(4-1)$$

$$B^* = 57.6 \text{ Gal/min}$$

Using the design specification for windage, 0.025% of the recirculating rate, or 3 gallons/minute, the calculated blowdown rate compares reasonably well with the measured rate of 52 gallons/minute.

Equation 1 is used to determine makeup requirements for the system and may be compared with other plant water consumption data (if available) for accuracy.

$$M = E + B^*$$

$$M = 172.8 + 57.6$$

$$M = 230.4 \text{ Gal/min}$$

To determine water treatment quantities needed to maintain a recommended concentration (115 ppm) in the recirculating water, use equation 4:

$$\text{lbs treatment/min} = \frac{M(\text{Gal/min}) \times \text{ppm (treatment)}}{C \times 120{,}000}$$

$$\text{lbs treatment/min} = \frac{\text{Gal/min} \times 115 \text{ ppm}}{4 \times 120{,}000}$$

$$\text{lbs treatment/min} = .0552 \text{ lbs/min} = 79.5 \text{ lbs/day}$$

Biocides are frequently slug fed to a cooling water system at a recommended dosage (65 ppm for example). The total system volume has been determined to be 75,000 gallons. The quantity of biocide required to slug dose the system can be determined from equation 5:

$$\text{lbs treatment} = \frac{\text{Volume(Gal)} \times \text{ppm (treatment)}}{120{,}000}$$

$$\text{lbs treatment} = \frac{75{,}000 \times 65}{(120{,}000)}$$

$$\text{lbs treatment} = 40.63 \text{ lbs}$$

Occasionally, the cooling water system becomes fouled with oil contamination. To determine how long it will take to blowdown 95% of a 300 ppm oil contamination at a maximum bleed rate of 100 gallons/minute use equation 6:

$$t \text{ (time)} = \frac{V \text{ (Gal)} \times \ln (C_i/C_f)}{B(\text{Gal/unit time})}$$

$$t \text{ (time)} = \frac{75{,}000 \text{ Gal} \times \ln (300/0.05 \times 300)}{100 \text{ Gal/min}}$$

$$t \text{ (time)} = \frac{75{,}000 \times \ln (20)}{100 \text{ Gal/min}}$$

In order to effect a complete kill at a biocide dosage of 65 ppm, it has been recommended that the cooling system have a half-life of at least 16 hours. To determine the half-life of the subject cooling system use equation 7:

$$t_{1/2} = \frac{0.69 \times V(\text{Gal})}{B^*(\text{Gal/unit time})}$$

$$t_{1/2} = \frac{0.69 \times 75{,}000 \text{ Gal}}{57.6 \text{ Gal/min}}$$

$$t_{1/2} = 898.4 \text{ min} = 14.97 \text{ hours}$$

Since the half-life of the system does not meet the recommended criteria, we may reduce the blowdown rate (if possible), increase the biocide dosage, shut off the blowdown for a short time period or change to a faster acting biocide.

Section V.
POTENTIAL PROBLEMS
CORROSION

Metal corrosion is a major concern in cooling water systems. The deterioration of metal surfaces caused by

---

* Includes windage (W), leaks, overflow losses, etc.

corrosion may result in corrosion product deposit accumulation on heat transfer surfaces and premature failure of equipment. Premature equipment failure will lead to process down time to replace the failing equipment and may call for a capital expenditure to purchase replacement equipment. Also, the deposition of corrosion products on heat exchanger surfaces reduces heat transfer efficiency and impedes water flow through the system.

The corrosion process consists of four basic components.

1. Anode - oxidation occurs at the anode and metal cations are produced.
2. Cathode - reduction occurs at the cathode; site where electrons leave the metal surface.
3. Electron Path - the electrons generated by the anodic reaction flow through the conductor (metal) to the cathode.
4. Electrolyte - provides necessary to complete the electrical circuit; electrolyte is needed for conducting positive charge from the anode to the cathode.

For example, iron is oxidized to divalent (ferrous) iron at the anode.

$$Fe \text{ ----->} Fe^{+2} + 2\ e^- \qquad (8)$$

Electrons flow from the anode, through the metal conductor to the cathode. In oxygenated water, oxygen is reduced to hydroxide ions at the cathode.

$$O_2 + 2\ H_2O + 4\ e^- \text{ ----->} 4\ OH^- \qquad (9)$$

The ferrous iron generated by the corrosion process is oxidized to ferric iron by oxygen dissolved in the water and the ferric iron precipitates as an insoluble hydrous oxide.

$4Fe^{+2} + O_2 + 80H^- + 2H_2O$ □□□□= $4Fe(OH)_3$(10)

The insoluble ferric hydroxide partially dehydrates to form the corrosion products (rust) commonly observed on steel surfaces.

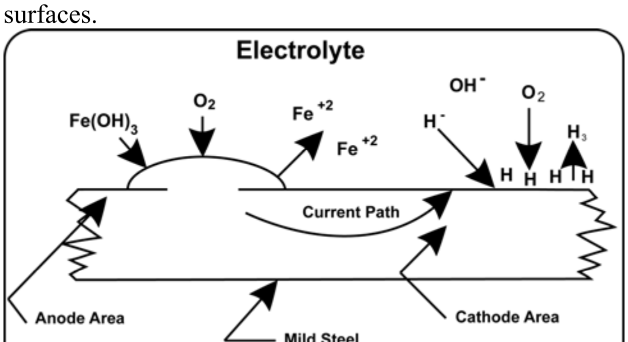

Figure 5
Schematic Representation Of Corrosion Process On Mild Steel In Water

Every metal surface is covered with anodic and cathodic sites. The anodic and cathodic sites arise from differences in the microstructure of the metal. Surface irregularities can develop during any metalworking process. These irregularities or differences create differences in surface energy or activity on the metal surface and encourage corrosion to begin at the more active sites.

For corrosion to proceed, the anodic (oxidation) and cathodic (reduction) corrosion reactions must occur simultaneously and at the same rate. Thus, corrosion can be inhibited by retarding either the anodic or cathodic reaction. The corrosion rate will be dictated by the slowest step. With cooling water operating in the near neutral to alkaline pH range, the rate determining step is typically oxygen reduction, which occurs at the cathode. The corrosion rate will therefore be limited by the rate at which oxygen diffuses to the cathode.

General Corrosion

Uniform metal loss from the surface is usually called general corrosion. General corrosion is the result of the constant shifting and interchange of the anodic and cathodic sites on the surface of the metal. The anodic site is not fixed, so corrosion occurs uniformly across the metal surface.

Pitting

Severe localized corrosion that results when the anode is fixed in one location is called pitting. Intense localized corrosion is the result of the anode being small compared to the cathode. The large cathodic reaction drives the anodic reaction. Typically pitting is defined as corrosion in which the depth of the metal loss is two or more times the width of the corroded area on the surface. Pitting corrosion is more damaging than general corrosion since it is leads to premature failure of metal walls.

Metals and alloys are not single, homogeneous phases, but may contain more than one chemical compound or phase. The different phases, when contacted with a conductive fluid, may result in a series of electrochemical cells. For example, brass metals contain a mixture of zinc and copper. The two elements are not perfectly mixed but contain very small crystal patches of copper and zinc thus setting up anodic and cathodic couples.

Carbon steel, an alloy of iron and carbon, is employed in water contact areas of most cooling water systems. Other metals and alloys used in the construction of cooling systems are copper, brass, bronze, stainless steel and galvanized steel. Copper is frequently used for heat exchanger tubes. Bronze (an alloy of copper and tin) is used for pumps, valves and other system equipment. Copper metal is relatively noble and does not corrode rapidly in the absence of aggressive ions and highly oxidizing compounds. Ammonia and oxygen together are aggressive toward copper and copper alloys as is chlorine.

The chrome-nickel stainless steels are very resistant to general corrosion by most aggressive ions encountered in open recirculating cooling systems, through the rapid formation of a thin protective iron oxide/chrome oxide film at the metal surface. However, these steels are subject to pitting type corrosion from differential aeration cell formation.

TABLE II

GALVANIC SERIES OF SELECTED METALS AND ALLOYS

Less Noble	Magnesium
(Anodic)	Zinc
	Galvanized Steel
	Aluminum 2S
	Cadmium
	Aluminum 17 St.
	Steel - Cast Iron
	Stainless (18-8) Active
	Solder (Lead, Tin)
	Inconel
	Monel, Cupronickel, Bronze, Copper, Brass
More Noble	Stainless (18-8) Passive
(Cathodic)	Titanium
	Gold
	Platinum

Another commonly encountered type of corrosion is <u>galvanic corrosion</u>. Galvanic corrosion occurs when two dissimilar metals are in electrical contact in a conductive medium. This still follows the basic rule that an anode, a cathode, a conductor and a complete electrical circuit are necessary for corrosion to occur. Metals and alloys, by their nature, have different relative electrical potentials. The galvanic series presented in Table II shows the relative potential of some commonly encountered metals. Metals and alloys near the bottom of Table II (more noble, or cathodic) are more stable in the zero valence (metallic) state. The less noble metals near the top of Table II have relatively low stability and are more easily oxidized. When two dissimilar metals come in contact with each other in a conductive liquid, the less noble metal becomes anodic to the more noble metal (cathode) and is oxidized. That is, the less noble metal corrodes. For example, when steel and copper come in contact in water, the iron becomes the anode and copper the cathode. Metal loss occurs at the iron surface as the iron corrodes.

<u>Under-Deposit Corrosion</u>

Corrosion occurring underneath deposits formed on the metal surface and shield portions of the surface from the circulating water is under-deposit corrosion. Dissolved oxygen is quickly consumed under the deposit and the metal surface under the deposit becomes anodic relative to the surrounding surface in contact with water. Pitting corrosion often begins in this manner.

<u>Crevice Corrosion</u>

This is a specialized form of under-deposit corrosion in which the anode develops in a crevice or stagnant area. Crevice corrosion is caused by the depletion of oxygen from the water in the crevice. This causes the metal in the crevice to be anodic compared to the metal adjacent to the crevice, which is exposed to oxygen saturated water. Crevice corrosion can be a problem with any metal, but it is especially important with aluminum and stainless steel. These materials depend on oxide layers for protection. The inhibiting oxide layers are stripped by the concentrated fluid in the crevice because of the lack of dissolved oxygen, leaving the metals exposed to corrosion.

Selective Leaching or Dealloying

The corrosion process in which one particular element of an alloy is removed is called selective leaching or dealloying. The most common example is dezincification of brass alloys, which are blends of copper and zinc. The metal structure appears sound because the remainder of the metal is intact, but the structure is weakened by the loss of alloying components. In cooling water systems, dealloying is often caused by low pH and stagnant conditions.

Erosion Corrosion

This is not an electrochemical form of metal degradation, but rather a mechanical problem created by excessive velocity or turbulence. Abrasive materials such as suspended solids and silt abrade the metal surface, removing previously formed oxide films. A new oxide film forms, only to be removed again. This process continues and significant amounts of material can be lost. Erosion can be controlled by reducing water velocities, removing suspended solids by filtration, and by coating susceptible parts such as pump impellers with a resistant material.

Stress Corrosion Cracking

Stress corrosion cracking requires a combination of tensile stress, an alloy metal and an environment that is specifically corrosive to that particular alloy. The failure does not necessarily follow grain boundaries in the metal. The most common causes of stress corrosion cracking are chloride ion cracking in austenitic stainless steels, caustic or hydrogen ion cracking of both stainless and carbon steels and ammonia cracking of brass alloys. Although most stress cracks originate from residual stresses, applied stresses originating from stretching or pressure on the metal can also cause stress corrosion cracking. The most reliable approach to solving this problem is to choose materials resistant to cracking and to ensure that all stressed components are properly stress relieved.

Microbiologically-Influenced Corrosion (MIC)

MIC is another special form of under-deposit corrosion. Anaerobic bacteria, specifically sulfate reducing bacteria (SRB) and acid producing bacteria (APB) can accumulate under-deposits. Metabolic reactions of these bacteria produce acids, dropping the pH low enough to cause serious localized or pitting attack. Almost any common alloy can be susceptible to MIC, but the problem is most prevalent in cooling water systems containing carbon steel, stainless steel and copper alloys. MIC often creates characteristic concentric rings on the pipe walls or coupon surfaces.

TABLE III	
Summary of Common Types of Scale Forming Minerals	
Scale	Chemical Formula
Acmite - Sodium Iron Silicate	$NaF_3Si_2O_6$
Barite - Barium Sulfate	$BaSO_4$
Analcite - Sodium Aluminum Silicate	$NaAl\ Si_2O_3 \cdot H_2O$
Aragonite - (Rhombic Crystals)	$CaCO_3$
Calcium Carbonate (Hexagonal Crystals)	$CaCO_3$
Calcium Sulfate - Anhydrite	$CaSO_4$
Hydromagnesite - Magnesium Carbonate and Hydroxide	$3MgCO_3\text{-}Mg\ (OH)_2 \cdot 3H_2O$
Hydroxyapatite - Calcium Phosphate	$Ca_5(PO_4)_3(OH)$
Iron Oxide	Alpha FeO(OH)
Iron Oxide - Magnetic	Fe_3O_4
Iron Oxide - Red	Fe_2O_3
Iron Chrome Spinels	$CrFe_2O_4$
Iron Sulfite - Trallite, Pyrrhotite	FeS
Manesium Hydroxide - Brucite	$Mg(OH)_2$
Magnesium Oxide - Magnesia	MgO
Manganese Dioxide - Pyrolusite	MnO_2
Montmorillonite - Aluminum Silicate	$Al2O_3 \cdot 4SiO_2 \cdot 4H_2O$
Noselite - Sidium Aluminum Silicate	$Na_8Al_6(SiO_4)_6SO_4$
Organic Deposits	
Pectolite - Calcium Sodium Silicate	$NaCa_2Si_3O_8OH$
Serpentine - Magnesium Silicate	$Mg_3Si_2O_7 \cdot 2H_2O$
Silica - Quartz	SiO_2
Sodalite - Sodium Aluminum Silicate	$Na_8Al_6(SiO_4)_6Cl_2$
Vermiculite - Magnesium Iron Aluminum Silicate	$(Mg,Fe)_6(Si,Al)_4O_{10}(OH)_2 \cdot H_2O$
Xonotlite - Calcium Silicate	$Ca_6Si_6O_{17}(OH)_2$

The compounds listed on the continuation of Table III are usually found in industrial equipment that contains brass or bronze:

TABLE III (Continued)	
Summary of Common Types of Scale Forming Minerals	
Scale	Chemical Formula
Copper Iron Sulfide	CuFeS
Copper Sulfide - Covelite, Chalcoclte	Cu_5 and Cu_2S
Basic Copper Choride	$CuCl_2 \cdot 3Cu(OH)_2$
Copper Oxide - Cuprite	Cu_2O
Chalcopyrlte	$CuFeS_2$
Beta Zinc Sulfide - Sphalerita	Zn_5
Green Basic Carbonate - Malachite	$CuCo_3Cu(OH)_2$

Scale Formation

Scale is a deposit that results from precipitation of dissolved solids from solution. When this precipitation occurs on heat transfer surfaces, serious losses of flow and heat transfer can occur.

Table III summarizes some of the scale-forming compounds encountered in the water treatment industry. Calcium carbonate and calcium phosphate are by far the most prevalent scales. Calcium sulfate scaling is not normally encountered, unless the calcium and sulfate concentrations are extremely high. Magnesium salts, including magnesium silicate, and silica can precipitate in high magnesium, high silica systems. Soluble iron entering with the makeup water (well water iron) or fresh iron generated from corrosion account for most iron oxide deposits. Copper salts can precipitate if corrosion of copper alloys is not controlled. Manganese in the makeup water can lead to manganese scale. Zinc deposits result from corrosion of galvanized steel or from improper use of zinc in corrosion control programs.

Scale Mechanisms

The potential for scale formation in open recirculating cooling systems is high for two reasons. First, the dissolved solids in the cooling water are concentrated due to the evaporation process. Each compound has a maximum solubility in water at any given temperature. When concentrated to its maximum solubility, the water is said to be saturated for that compound. Further concentration will result in precipitation of the compound from solution.

Second, many of the scale forming compounds encountered in recirculating cooling systems have solubility-temperature relationships different from that conventionally expected. The conventional solubility-temperature relationship for a compound in solution is increasing solubility with increasing temperature (see Figure 6). However, many of the scale forming compounds found in open recirculating cooling systems do not have conventional solubility-temperature relationships. In the temperature range of typical circulating cooling systems, these compounds become less soluble with increasing temperature (sometimes called inverse or retrograde solubility). The hottest areas of the cooling system are the heat exchanger surfaces, and these are the most likely sites for precipitation of inversely soluble compounds.

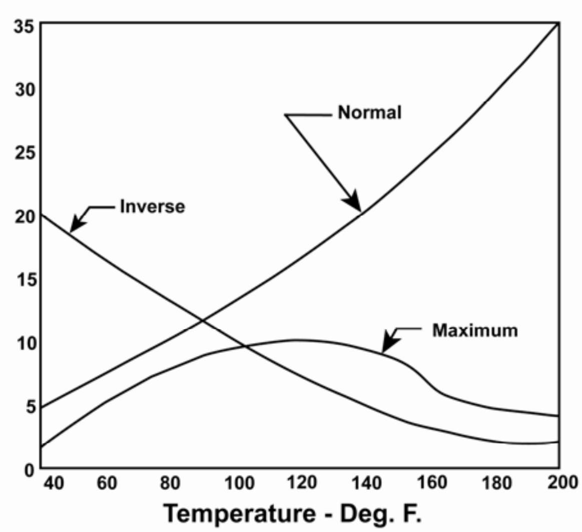

Figure 6
Effects of Temperature On Solubility

The scale resulting from the precipitation of inorganic salts generally has a high resistance to heat transfer. That is, scale is an effective insulator that reduces the efficiency of heat transfer within the process equipment. In extreme cases, where the thickness of the deposit is significant, water flow restrictions may occur. If the scale sloughs from the surfaces, it may collect at other points in the system. Scale of any kind is a deposit, and can create conditions suitable for under-deposit corrosion. (See the corrosion portion of this section for more on under-deposit corrosion.)

The solubility limit for any chemical compound may be defined in terms of is solubility product constant, K_{SP}. For example, the solubility product for a compound A_X B_Y would be written as:

$$\underline{[A]^X [B]^Y} = K_{SP} \qquad (11)$$

Where <u>[A] and [B]</u> are the <u>molar</u> concentrations in solution of A and B ions, respectively. When the product of [A] times [B] exceeds the solubility product constant, insoluble A_X B_Y will result. Simply stated, the solubility product constant may be exceeded by increasing the concentrations of A and/or B.

Some scale forming compounds precipitate as dense deposits with well defined crystal structures. These compounds are easily identified by modern analytical instrumentation. Other scale forming compounds precipitate as amorphous compounds with poorly defined crystal structures. Even with modern analytical instrumentation, it is sometimes difficult to distinguish these amorphous scale compounds from other water borne suspended solids.

Following is a more detailed discussion of some of the most common scale forming compounds.

<u>Calcium Carbonate</u>

Insoluble calcium carbonate forms when the concentrations of calcium (Ca^{+2}) and carbonate ($CO3^{-2}$) ions exceed the solubility of calcium carbonate.

$$Ca^{+2} + CO_3^{-2} \text{ -------> } CaCO_3 \qquad (12)$$

Most cooling water makeup supplies contain appreciableconcentrations of calcium and bicarbonate (HCO_3^-) ions. The concentration of carbonate ion in the makeup is normally low.

A chemical equilibrium exists between the concentrations of carbonate ion, bicarbonate ion and carbonic acid in solution. (Carbonic acid results from carbon dioxide in the air dissolving in the cooling water.) The equilibriums can be expressed as:

$$CO_3^{-2} + 2H^+ \text{<-->} H^+ + HCO_3^{-1} \text{<-->} H_2CO_3 \text{<-->} CO_2 + H_2O \qquad (13)$$

High pH shifts this equilibrium towards the left (formation of carbonate) and low pH shifts the equilibrium towards the right (formation of carbonic acid and carbon dioxide).

In an open recirculating cooling system air contacting the water in the cooling tower tends to scrub carbon dioxide from the water. The result is a shift of a portion of the bicarbonate ion to carbonate ion.

$$2\ HCO_3^{-2}1 \text{------->} CO_3^{-2} + CO_2 + H_2O \qquad (14)$$

Heat decomposes bicarbonate to form carbonate ions by this same reaction. The net result of the heat input into the cooling water is to increase the carbonate ion concentration, both by conversion from the bicarbonate ion and by concentration due to evaporation. The concentrating effect due to evaporation also increases the calcium ion concentration in the cooling water. Consequently, the saturation limit for calcium carbonate is exceeded, at low cycles of concentration, in most open recirculating cooling systems.

Calcium carbonate may exist in any of three different crystal structures. The type which normally forms in cooling systems is calcite, although aragonite and vaterite have been reported. Calcite has an inverse solubility temperature relationship (Figure 7) and forms with a well defined crystal structure. It normally occurs on heat exchange surfaces as a hard, dense, insulating scale (Figure 8). In severe cases, calcite scale may also form in other areas of cooling systems such as on the cooling tower fill. Calcium carbonate is white but may appear cream-colored due to small amounts of impurities , e.g., iron salts.

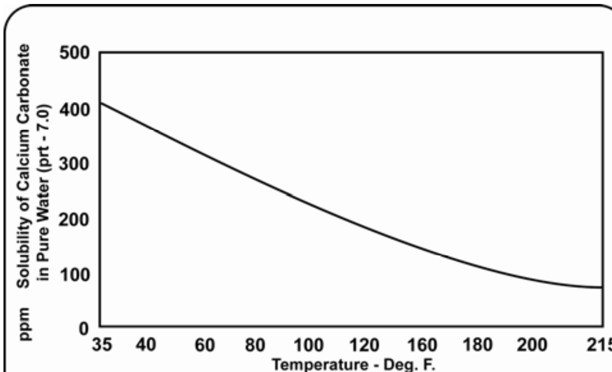

Figure 7
Solubility Of Calcium Carbonate In Water

An easy test for the presence of calcium carbonate is to add a small quantity of acid to the scale. If the deposit effervesces strongly, the deposit contains carbonate and probably is calcium carbonate. The bubbles are carbon dioxide which are released from the carbonate ion as shown in equation (13) above.

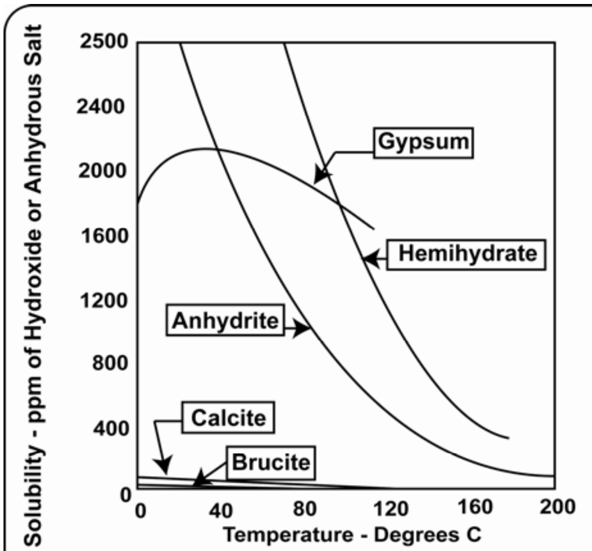

Figure 8
Solubility of Scaling Compounds In Water

<u>Calcium Sulfate</u>

Calcium sulfate forms when the concentrations of calcium and sulfate ions in the water exceed the solubility product of calcium sulfate.

$$Ca^{+2} + SO_4^{-2} \text{ -------> } CaSO_4 \qquad (15)$$

Calcium sulfate is considerably more soluble than calcium carbonate. Consequently, calcium sulfate solubility is exceeded in a comparatively small number of recirculating cooling systems. Systems using makeup water containing high calcium and/or high sulfate concentrations may have the potential for calcium sulfate scale formation. Systems using high bicarbonate and high calcium makeup water may employ sulfuric acid to avoid calcium carbonate scale formation. This may result in sulfate ion concentrations sufficient to exceed the solubility of calcium sulfate.

Calcium sulfate exists in three different crystal structures (see Figure 8). Gypsum is the type that most often forms in the heat exchange equipment in recirculating cooling water systems. Another crystal type, anhydrite, can form in high temperature heat exchangers. Both gypsum and anhydrite have inverse solubility relationships. Both crystal types have well defined, easily identifiable crystal structures and form dense, hard scale on heat exchange surfaces. It is extremely difficult to remove calcium sulfate scale with any of the common chemical cleaning agents.

<u>Calcium Phosphate</u>

Orthophosphate ion concentrations (PO_4^{-3}) sufficient to create a potential for calcium phosphate scale formation may result from one or more sources. Orthophosphate is widely used as a steel corrosion inhibitor at both near neutral and alkaline ph conditions. Makeup waters containing either agricultural runoff or domestic sewage may contain sufficient phosphate to form calcium phosphate scale when the water is concentrated in the cooling tower. Water treatment chemicals added to the recirculating water for corrosion and/or scale control frequently contain phosphorous compounds that may be converted to orthophosphate in cooling systems.

Three classes of phosphorous compounds are used in cooling water systems: inorganic ortho and polyphosphates and organic phosphorous-containing compounds. Inorganic polyphosphates hydrolyze (react with water) and can form significant concentrations of orthophosphate in relatively short time periods. The potential for calcium phosphate scale formation is always high when phosphate-based corrosion inhibitors are used. Also, many corrosion and scale inhibitors contain organic phosphorous compounds. These compounds may be oxidized and form orthophosphate ions in the recirculating water.

A chemical equilibrium exists between the various phosphate ion species and the hydrogen ion. This can be illustrated by the dissociation of phosphoric acid:

$$H_3PO_4 \text{ <--> } H_2PO_4^{-} + H^{+} \text{<-->} HPO_4^{-2} + 2H+ \text{<-->} PO_4^{-3} + 3H^{+} \qquad (16)$$

As the hydrogen ion concentration is increased (pH decreased), other conditions remaining constant, the equilibrium in the equation is shifted from the right toward the left. As the pH is increased, hydrogen ions are consumed and the equilibrium is shifted toward the right, increasing the concentration of orthophosphate ions. Thus, calcium phosphate solubility is pH dependent; decreasing with increasing pH.

Calcium phosphate precipitates in recirculating cooling systems as one or more compounds with poorly defined crystal structure. Calcium phosphate frequently forms in combination with calcium carbonate scale and water-born suspended solids. Consequently, it is sometimes difficult to determine the extent to which heat exchanger fouling is due to calcium phosphate scale formation. Calcium phosphate scale is less tenacious than calcium carbonate and can be removal can be accomplished by both mechanical and chemical methods.

<u>Silica and Metal Silicate Scales</u>

Insoluble silica scale can result from the polymerization of silica (SiO_2) in open recirculating cooling systems. This scale is a very hard, glassy, amorphous (non-crystalline) material. Polymerized silica has a conventional solubility-temperature relationship and consequently tends to precipitate on the lower temperature surfaces of the system. Other conditions being constant, silica scale potential increases with decreasing cooling water pH, because lower pH encourages silica polymerization.

The silicate ion (SiO_3^{-2}) may react with any of several metal ions in the recirculating cooling water to form insoluble metal silicates. Among the metal silicates sometimes observed are aluminum silicate, magnesium silicate and zinc silicate. These compounds usually precipitate as amorphous materials and are difficult to define using modern instrumental deposit analysis techniques. Consequently, it is frequently hard to distinguish these scales from deposited water borne suspended matter.

The potential for formation of metal silicate scales is greater in high pH cooling water. These compounds appear to have inverse solubility-temperature relationships. The metal silicate scales, and especially silica scale, are very difficult to remove by chemical cleaning techniques.

<u>Saturation Indices and Critical Temperatures</u>

As explained above, scaling occurs in recirculating cooling water when the concentration of specific ions, e.g. calcium and carbonate, exceed the solubility product of a specific scale-forming compound, e.g. calcium carbonate. Solubility products vary with temperature, pH and the ionic strength (conductivity or total dissolved solids).

Various empirical methods have been developed to predict whether or not scale formation will occur in a given recirculating cooling water system. Perhaps the simplest of these is a practical limit on the solubility product for calcium sulfate. Experience has shown that if the solubility product (expressed as calcium carbonate) is less than 500,000, calcium sulfate scale is not likely to form. That is, to avoid calcium sulfate scaling the concentrations of calcium and sulfate ions should meet the requirement of equation 17:

$$(Ca^{+2}) \times (SO_4^{-2}) < 500{,}000 \qquad (17)$$

Where (Ca) is expressed as ppm $CaCO_3$ and (SO_4) is expressed as ppm SO_4.

The well-known Langelier Saturation Index (LSI) was developed to predict the tendency of calcium carbonate to precipitate. The LSI depends upon an empirical calculation of the pH (pH_S) at which calcium carbonate will become saturated in a given solution. pH_S is a function of temperature, total dissolved solids, calcium and total alkalinity, as shown in Table IV. pH_S is calculated by adding the factors for each of these parameters. The LSI is then defined as the difference between the actual pH of the water and pH_S. That is:

A positive LSI indicates that calcium carbonate will precipitate, and a negative LSI indicates that calcium carbonate scale will dissolve. It must be remembered that the LSI predicts only a tendency for precipitation or dissolution to occur. It does not indicate that calcium carbonate actually will form, or if it does, how fast or to what extent precipitation will occur.

Table V shows a sample of LSI calculation based on the data in Table IV.

TABLE IV							
Data for Rapid Caculations of the Langilier Index							
(Calcium Carbonate Saturation Index)							
(Based on the Langelier Formula, Larson-Buswell Residue, Temperature Adjustments: Arranged by Nordell)							
Total Solids	A	Temperature (^oF)	B	Calcium Hardness (ppm of $CaCO_3$)	C	Alkalinity (ppm of $CaCO_3$)	D
50-300	0.1	32 - 34	2.6	10 - 11	0.6	10 - 11	1.0
400-1000	0.2	36 - 42	2.5	12 - 13	0.7	12 - 13	1.1
		44 - 48	2.4	14 - 17	0.8	14 - 17	1.2
		50 - 56	2.3	18 - 22	0.9	18 - 22	1.3
		58 - 62	2.2	23 - 27	1.0	23 - 27	1.4
		64 - 70	2.1	28 - 34	1.1	28 - 35	1.5
		72 - 80	2.0	35 - 43	1.2	36 - 44	1.6
		82 - 88	1.9	44 - 55	1.3	45 - 55	1.7
		90 - 98	1.8	56 - 69	1.4	56 - 69	1.8
		100 - 110	1.7	70 - 87	1.5	70 - 88	1.9
		112 - 122	1.6	88 - 110	1.6	89 - 110	2.0
		124 - 132	1.5	111 - 138	1.7	111 - 139	2.1
		134 - 146	1.4	139 - 174	1.8	140 - 176	2.2
		148 - 160	1.3	175 - 220	1.9	177 - 220	2.3
		162 - 178	1.2	230 - 270	2.0	230 - 270	2.4
				280 - 340	2.1	280 - 350	2.5
				350 - 430	2.2	360 - 440	2.6
				440 - 550	2.3	450 - 550	2.7
				560 - 690	2.4	560 - 690	2.8
				700 - 870	2.5	700 - 880	2.9
				880 - 1000	2.6	890 - 1000	3.0

TABLE V
SAMPLE SCALING INDEX CALCULATIONS

	Scaling	Non-Scaling
1. Langelier Saturation Index	Positive Value	Negative Value
2. Ryzner Saturation Index	<6	>6

Example 1 Basis pH = 8 at 2 cycles of concentration

Critical temperature, 120°F

TDS = 210

Ca as $CaCO_3$ = 120

M Alk as $CaCO_3$ = 100

Therefore, from Table IV A = 0.1

B = 1.6

C = 1.7

D = 2.0

pHs = (9.3 + A + B) – (C + D)

pHs = (9.3 + 0.1 + 1.6) – (1.7 + 2.0) = 7.3

L.S.I. = 8 – 7.3 = +0.7

Another calcium carbonate saturation index in common industrial use was devised by Ryznar. The Ryznar Saturation Index does not depend purely on empirical data, but tries to correlate actual operating experience with empirical equations. The Ryznar Index is defined as $2pH_S$ - pH, with the nominal neutral point at 6.5. Values below 6.0 are scale-forming while those above 7.0 are corrosive. The range of 6.0 to 7.0 is considered neutral, or stable.

Saturation pH values for calcium phosphate and calcium sulfate scales can be calculated by methods similar to that used for pH_S for calcium carbonate. Data for making these calculations are readily available in the water treatment handbooks.

Predictive capabilities can be further improved when a computer is used to calculate some of the variables associated with supersaturation. For example, total ionic strength, alkalinity redistribution, specific ion activity, and formation of ion pairs all have significant effects upon the solubilities of scale forming compounds. Computer programs are available that include these parameters in the pH_S calculations.

It should also be pointed out that all saturation predictions are based on some specified temperature (variable B in Table IV). This temperature can either be that of the bulk liquid, the scale liquid interface or the wall temperature. For most applications, where a scale with inverse solubility, such as calcium carbonate, is involved, the hottest temperature in the system should be selected. This is the exit temperature from the hottest heat exchanger **plus** a 25°F correction factor. This factor is used as a guesstimate to predict the unmeasured skin (surface) temperature. In a practical sense, if heat transfer surfaces are kept clean at high skin temperatures, then all other heat exchangers operating at lower temperatures should also remain scale-free.

Growth of Microbiological Organisms

The water and wetted areas of open recirculating cooling systems present an environment in which several types of microbiological organisms can grow and propagate. Microbiological organisms enter the system with the makeup water and with airborne particulates scrubbed in the cooling tower. The classes of microbiological organisms that proliferate in open recirculating cooling systems are algae, fungi (yeast and mold) and bacteria.

Algae

Algae require sunlight and use chlorophyll to convert carbon dioxide into biomass. While different types of algae flourish under different conditions, those which are most predominant in recirculating cooling systems, are found on the wet, exposed, aerated surfaces such as the open distribution decks of the cooling tower. Algae biomasses are found attached to the tower deck, and may also be found on tower structural members and in the plenum and mist eliminators. Algae attach themselves to structural surfaces in the area where they grow. Consequently, algae normally have little direct effect on heat exchanger surfaces. If dislodged, algae may interfere with proper water distribution on the tower deck and/or they may be transported to heat exchangers where they may cause plugging.

Algae biomass can provide a nutrient source for bacteria, thereby enhancing bacterial growth. If attached to, and/or deposited on metal surfaces, algae can contribute to localized corrosion processes, especially microbiologically induced corrosion (MIC).

Fungi

Fungi are simple organisms containing no chlorophyll. They can be unicellular or filamentous. They usually require less moisture and can survive at lower pH levels than algae or bacteria. Fungi can reproduce both sexually or asexually (sporulation). Two commonly known classifications of fungi are yeasts and molds. The most serious damage caused by molds is destruction of cooling tower wood. Fungi obtain their food from plant and animal matter by secreting enzymes into their surroundings. Cellulytic fungi use cellulose as a source of carbon, and in doing so, they destroy the wood. The most generally accepted classifications of fungal wood destruction are soft rot (sometimes called surface rot), brown rot and white rot (sometimes called pocket rot or deep rot).

Soft rot occurs primarily on wood surfaces that are heavily wetted, such as the tower fill. The cellulose (the material giving the rigidity to wood) is destroyed, while the lignin (the cell cementing material in wood) is not significantly attacked. Wood which has suffered soft rot, upon drying, will reveal cracks perpendicular to the grain, giving a cross-checked appearance to the wood. Also, the wood becomes very brittle; if broken across the grain, the wood will not splinter, but will break evenly in a straight line along the edge.

Brown rot is similar to soft rot, but it occurs inside the wood. Again, the cellulose is metabolized, leaving the lignin little affected. A brown color occurs as a result of the lignin residue remaining. Brown rot occurs in wood that is not fully saturated with water, allowing diffusion of air into the wood. Areas of the tower, such as the plenum, that are only contacted with water mist, are more likely to incur brown rot. Wood may suffer brown rot and lose most of its structural strength, yet it may appear, externally, to be sound.

White rot also occurs inside wood not completely saturated with water. White rot organisms digest both the cellulose and the lignin, leaving hollow pockets in the wood. Wood that has suffered white rot may also appear sound when viewed externally.

The yeast type of fungi do not cause wood rot, but instead can proliferate to high numbers and foul heat exchanger surfaces.

Bacteria

Bacteria are single cell, microscopic organisms that usually reproduce by binary fission. Bacteria can be described as aerobic, anaerobic or facultative, depending on whether they flourish in an oxygenated environment, an environment void of oxygen, or in both types of environments, respectively.

One of the more predominant types of bacteria occurring in open recirculating cooling water systems are slime-forming bacteria. These bacteria are aerobes and are naturally present in the soil. Consequently, they continuously infect open recirculating cooling systems. These bacteria may flourish throughout the cooling system, including the heat exchanger surfaces and the cooling tower (see Figure 10). These are typically encapsulated bacteria that produce slime layers outside the cell walls. These slime-encased cells attach to the available surfaces. Then they reproduce, form additional slime and develop into biodeposits or "biofilms". The biofilm layer is composed mostly of water, hence it presents a significant barrier to heat exchange. The biofilm mass, which also frequently contains filamentous bacteria, can serve as the nucleus for agglomeration and deposition of water borne suspended solids. These combined biological-mineral deposits dramatically reduce heat exchanger efficiency and also create differential aeration cells that can result in high localized corrosion rates.

Corrosion of metal surfaces is also caused by several types of anaerobic bacteria. These bacteria flourish beneath biofilm and other deposit layers where oxygen is not present or readily replenished. The most notable of the anaerobic corrosive bacteria are the sulfate reducing bacteria (SRB). These bacteria reduce sulfate to corrosive hydrogen sulfide. They also are thought to cause cathodic depolarization by removal of hydrogen from the cathodic portion of corrosion cells. Acid producing bacteria (APB) produce organic acids. These metabolic processes cause localized corrosion of deposit laden distribution piping and also provide the potential for severe pitting corrosion of heat exchanger surfaces. This entire process is called microbiologically induced corrosion, or MIC.

Legionella bacteria, which can pose potential health hazards, can also proliferate in cooling towers and condensers. Legionella bacteria are normally found in low numbers in water-containing systems and can be isolated from most natural aquatic and soil environments. Cooling towers are therefore easily contaminated with these bacteria. When Legionella are expelled from towers as an aerosol in the drift, there is the potential that people may contract a certain type of pneumonia called Legionnaires' Disease or Legionellosis. Although people inhaling these bacteria-containing aerosols may get severe pneumonia and require antibiotic therapy in a hospital, the conditions necessary to produce the disease from cooling tower aerosols are certainly not well established. A less serious form of the disease is called Pontiac Fever. Legionella bacteria are not slime-forming and do not appear to cause fouling or corrosion in cooling towers.

Fouling

One of the fundamental problems facing the water treatment industry today is the persistence of "fouled" heat transfer surfaces, inlet screens, pipes, etc. The term "fouling" means the deposition of any undesirable material in the cooling water system. Fouling may occur on heat exchanger tubes reducing effective heat transfer (Figure 11), or it may occur on the tower deck and fill, interfering with water flow and cooling efficiency.

Sources of Fouling Substance

Fouling, as a general term, covers deposition of any type of precipitated or suspended matter in the system. This can include, among other things, corrosion products, biological growths, mineral scales and water-borne sediments.

Corrosion, scale and microbial deposits are discussed in Sections V., A., B. and C. above. Each of these phenomena produce an insoluble material that can travel through the system and deposit on heat transfer surfaces and in the cooling tower.

Another biological growth problem in cooling water systems results from macrobiological fouling. Marine organisms, such as clams and mussels, enter the cooling system in the larval stage. Since these larvae are very small, they easily slip by the plants' screening devices. Once inside the cooling system, they proliferate and can completely block heat transfer tubes. These organisms are difficult to kill because they close tightly at the first sign of chlorine. Other organisms such as sponges clog inlet screens.

Sedimentation

Water-borne suspended matter typically enters the cooling system with the makeup water or the air. The most common type of makeup water contaminants are silt and other finely divided soil particles.

Soil particles typically contain clay. Common types of clay material that have been identified in cooling water deposits include kaolinite and montmorrillonite. Clays are predominantly aluminum silicates of varying ratios, with differing amounts of other constituents. Differing mixtures of clays are found in various geographic areas. These clays enter the cooling water with river water makeup, and with air-blown solids. Clay-type deposits can entrap microbes and serve as a substrate for biological fouling.

If the makeup water is clarified to remove suspended solids, the makeup may be contaminated with primary coagulants such as aluminum hydroxide or ferric hydroxides. In addition, precipitants such as calcium carbonate or magnesium hydroxide may appear because of carryover from lime-soda softening processes. In addition to clays, air scrubbed particulates include silt, sand, organic matter and biological contaminants. Potential process contaminants are too numerous to discuss. Among the more troublesome deposit-forming process contaminants are soluble hydrocarbons and insoluble oils.

The potential for deposition of iron and manganese oxides is a problem in many recirculating cooling systems using well water makeup. These metals exist in reduced valence states in the well water and are oxidized to higher valence states when exposed to air. Both iron and manganese compounds are soluble in the reduced valence state, but because of their low solubility when oxidized by air, the oxides form suspended solids in the system.

Suspended metal oxides may also be present because of ongoing corrosion in the cooling water system. Areas of low water velocity permit these and other transportable suspended solids to settle and become sites for secondary under-deposit localized corrosion. This process is sometimes called deposit induced corrosion.

The effects that sedimentation fouling have on system integrity are difficult to predict. What is known is that suspended solids are constantly being transported from point to point within the system. If they are kept moving at a high enough velocity, they will not settle out and impede heat transfer or generate corrosion cells. The use of dispersants also helps to keep suspended solids from settling.

Removal of suspended solids from the makeup water and, if needed, sidestream filtration of the recirculating water, are the most reliable methods available for controlling sedimentation fouling problems in open recirculating cooling water systems.

Section VI.
Chemical Treatment for Corrosion, Scale and Deposit Control

Many different kinds of chemical treatment are used to minimize problems and ensure efficient and reliable operation of cooling systems. Each open recirculating cooling system is unique. Among those characteristics that make one system different from another are:

- System design, including size, basin depth, materials of construction, flow rates, heat transfer rates, temperature drop and other factors.
- Water, including makeup water composition, pretreatment and cycles of concentration.
- Contaminants, including process leaks and airborne debris.
- Wastewater discharge restrictions.

The selection of a treatment program for a specific system must take into account all of these system characteristics and others, and must contain materials to control, to an acceptable degree, each of the potential problems that may occur in that system.

It is fortuitous that many of the chemicals used to treat cooling systems are effective in controlling more than one problem. For example, some carbon steel corrosion inhibitors also effectively control calcium carbonate scale.

Corrosion Inhibition

Materials of construction used in cooling systems are selected for their physical, mechanical and thermal properties and for their resistance to deterioration in the operating environment. However, compromises are often used in recirculating cooling systems, particularly in the cooling tower itself. These materials may require operating and/or chemical restrictions to control their deterioration. Carbon steel is commonly used because of its favorable cost and good mechanical properties. Many other alloys are employed where their mechanical and thermal properties, and/or their corrosion resistance are required.

Corrosion inhibitors are almost universally used to control deterioration of steel and copper alloys in cooling systems. Corrosion inhibitors work by interfering with chemical reactions of metals with the aqueous environment. They may restrict either the anodic or the cathodic reaction. Inhibitors that restrict reactions at the anodes are called anodic inhibitors and inhibitors that work at the cathodes are called cathodic inhibitors. Some corrosion inhibitors impede both the anodic and cathodic reactions. These are called mixed inhibitors.

Many different types of chemicals are employed as corrosion inhibitors. Only those inhibitors commonly used in cooling systems are discussed here. The effectiveness of any specific inhibitor composition, and the quantities required for good results, depend upon the conditions in each specific system, and are beyond the scope of this discussion.

Chemicals used as corrosion inhibitors will be discussed here primarily as individual compounds. Most practical corrosion inhibitor programs employ combinations of two or more components. Anodic and cathodic inhibitors are often used together to establish control over both sides of the corrosion reaction. Such mixtures are said to be synergistic when the performance of the mixture is seen to be greater than the performance expected from the individual components added together. Because of this effect, some inhibitors that are known to be weak when used alone, can make important, cost-effective contributions to corrosion control when used in combinations with other inhibitors. Some typical combination inhibitors are discussed later in this section.

<u>Corrosion Inhibitors for Carbon Steel</u>

Carbon steel, an alloy of iron with small amounts of carbon, corrodes by reacting with the dissolved oxygen in open cooling systems. In most systems, some form of chemical treatment of the water is required to control this corrosion. The most widely-used corrosion inhibitors for carbon steel are discussed below.

<u>Polyphosphate</u>

One of the most common corrosion inhibitors for carbon steel in open recirculating cooling systems is polyphosphate. Polyphosphates are inorganic phosphate polymers prepared by dehydrating phosphoric acid or simple phosphate salts. They may exist either as straight chain or cyclic polymers. Many different names are used to describe these polymers, including; molecularly dehydrated, condensed polymeric, poly and meta phosphates. Van Wasser described all polymerized phosphates as condensed phosphates. He assigned the term metaphosphates to straight chain polymers and polyphosphates to those with ring structures. Following common practice in the water treatment field, the term polyphosphate is used here to include all dehydrated, polymeric inorganic phosphates.

Polyphosphates are effective cathodic corrosion inhibitors for carbon steel. They form either calcium or heavy metal polyphosphate films on the cathodic surfaces of the metal. Therefore, the cooling water must contain appreciable amounts of calcium or a metal ion such as zinc, in order to see good performance from polyphosphates.

Polyphosphates can provide good corrosion protection over a pH range from 6.5 to above 8.5. Polyphosphates begin to hydrolyze (react with water) and to form orthophosphate ions immediately on dissolving in water. The rate of this reversion reaction increases with lower pH and higher temperatures. Certain bacterial enzymes also can increase the reversion rate of polyphosphates. This formation of orthophosphate often leads to precipitation of calcium phosphate in systems containing appreciable amounts of calcium. For this reason, calcium phosphate scale inhibitors are almost always included in phosphate based corrosion inhibitor programs for open recirculating cooling systems. Polyphosphate reversion is not affected by oxidizing microbiocides in the cooling system. Polyphosphates also inhibit calcium carbonate scale (see below).

Orthophosphate

The orthophosphate ion (PO_4^{-3}) and its partially protonated forms (HPO_4^{-2} and $H_2PO_4^{-1}$) are anodic corrosion inhibitors for carbon steel. Anodic inhibitors promote the formation of a gamma iron oxide film at the anode. Since the orthophosphate ion is not an oxidizing agent, dissolved oxygen is needed in the water to form the oxide film. If the system is not well controlled, so that some corrosion occurs, deposits of iron phosphate can form at the anodes. These deposits are not protective and can actually encourage under-deposit corrosion.

Orthophosphates may be added directly to the cooling water, or they may be formed from hydrolysis of polyphosphates. As explained above, phosphate-based programs, particularly those containing orthophosphates, almost always include a calcium phosphate scale inhibitor.

Chromate

Chromate, that is, the hexavalent chromium anion (CrO_4^{-2}), is a very effective carbon steel corrosion inhibitor. At concentrations greater than about 150 ppm, chromate acts as an anodic inhibitor. Since it is a strong oxidizing agent, chromate does not require dissolved oxygen for generation of the protective gamma iron oxide film at the anodes.

At lower dosages; e.g., 10-25 ppm CrO_4^{-2}, chromate shows both cathodic and anodic properties. It is believed to perform as a cathodic inhibitor by accepting electrons and being reduced to trivalent chromium ($Cr^{+}3$) at the cathodes. This results in the formation of a good protective film of chromium oxide at the cathodic surfaces.

Chromate is effective over a wide pH range, from 6.0 to above 9.5, and chromate is not affected by oxidizing microbiocides. Chromate does not act as either a scale or a suspended solids deposit inhibitor. Chemical reducing agents, such as hydrogen sulfide and some hydrocarbons, can reduce chromate to trivalent chromium, which has very limited solubility in cooling water. Thus contamination with reducing agents may result in loss of chromate and deposition of trivalent chromium.

Both chromate and trivalent chromium are classified as toxic substances, and their discharge is essentially banned by federal regulation is the U.S. However, they are mentioned here since the use of low levels of chromate, generally with zinc salts has proven to be the standard to which most current treatments are compared. In addition, there are parts of the world where chromates may still be used. In cases where chromate and other heavy metals may not be environmentally controlled, a chromate/zinc salt treatment program is probably the most effective available.

Zinc

Zinc, by itself, is a weak inhibitor of carbon steel corrosion. However, when zinc is combined with other inhibitors (see below), it can substantially improve overall corrosion protection. Zinc acts at the cathodes, by forming a film containing either zinc hydroxide or another sparingly soluble zinc salt.

When used by itself, zinc is restricted to the near neutral pH range. Above approximately pH 7.5, zinc hydroxide precipitates from solution as the hydroxide or as various organic zinc salts. When this happens, corrosion inhibition is lost and the zinc may deposit on heat transfer surfaces. The useful pH range may be extended upward by including stabilizers (complexing agents) or inhibitors to prevent zinc precipitation. The zinc concentration in discharged wastewater is regulated and has essentially been highly limited in recirculating cooling tower systems in the United States. However, in other parts of the world zinc remains a viable treatment option. In general, zinc salts should be used with other treatment options such as chromate and phosphate.

Organic Phosphorous Compounds

Organic phosphorous compounds contain either direct carbon-phosphorous bonds (phosphonates) or carbon-oxygen phosphorous bonds (phosphate esters). These compounds are active both as calcium carbonate scale inhibitors (see below) and as corrosion inhibitors for carbon steel. The phosphonates in particular show weak to strong corrosion inhibiting properties, depending on the particular phosphonate, the water composition and the operating conditions in each system.

Organic phosphorous compounds appear to act primarily as cathodic inhibitors, but they may also have some direct effect on the anodic reaction. These compounds require either calcium or a metal ion, such as zinc, for effective corrosion inhibition, although some of the new phosphonates seem to perform well in very soft, low hardness cooling water.

Molybdate

The molybdate ion (MoO_4^{-2}) acts as an anodic corrosion inhibitor for carbon steel. Molybdate is a weak oxidizing agent, so that dissolved oxygen in the water is required for good performance unless very high levels of molybdate (above about 100 ppm) are used.

Molybdenum and chromium appear in the same column of the periodic table of the elements and molybdate and chromate have parallel chemical structures. It is sometimes assumed, therefore, that molybdate will show the same level of corrosion inhibition for carbon steel as chromate, with the advantage that molybdate is a far less toxic material. This, however, is not the case. Molybdate is a weak inhibitor. If used alone, very high levels are required for good results. Alternatively, molybdate can be used with other cathodic and anodic corrosion inhibitors to give synergized performance as described below.

Combination Inhibitors

When used alone, many of the corrosion inhibitors described above may not be cost effective or may not provide the required level of protection. However, mixtures of corrosion inhibitors are often synergistic and can provide excellent protection at costs far lower than would be required for individual components. Following are some of the combination corrosion inhibitors often used in open recirculating cooling systems:

Chromate-Zinc	Phosphate-Phosphonate
Phosphonate-Zinc	Phosphate-Polyphosphate
Phosphate-Zinc	Phosphonate-Molybdate
Carboxylate-Zinc	Phosphate-Molybdate
Molybdate-Zinc	

Other combination inhibitors with demonstrated cost-effective performance also are used. Many of these combination products are protected by patents.

Combinations of inhibitors in all proportions may not always provide cost-effective synergism. Some inhibitor combinations provide good results over a broad range of relative dosages. In other cases, the relative proportions that provide cost-effective synergism may be much more limited. This is illustrated by the chart in Figure 9 showing the performance of orthophosphate-polyphosphate mixtures.

In this test, when either orthophosphate or polyphosphate was used alone at a dosage of 15 mg/l as $PO^{4,}$ the carbon steel corrosion rate was approximately 1.8 mpy. When the two inhibitors were mixed at 60% orthophosphate and 40% polyphosphate while keeping the total phosphate concentration constant at 15 mg/l, the corrosion rate was approximately 0.3 mpy. This represents an 83% reduction in corrosion rate, obtained by combining these two inhibitors at their optimum synergistic ratio. Greater than 70% reduction in corrosion rate was obtained over the entire range of relative dosages, as long as each inhibitor was present at greater than 20% of the total concentration.

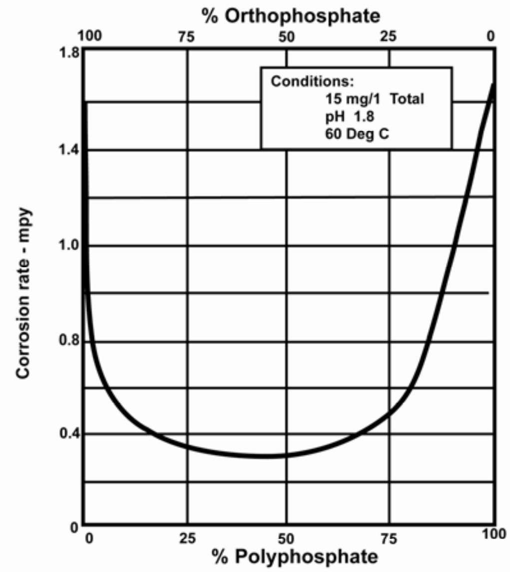

Figure 9
Corrosion Inhibition With Orthophosphate Polyphosphate Mixture

Nitrite and Silicates

The nitrite ion (NO_2^{-1}) is an anodic corrosion inhibitor that promotes formation of gamma iron oxide films. It is an oxidizing agent and does not require oxygen to form the protective oxide film. Nitrite is a common component of closed recirculating cooling systems, where it can provide excellent corrosion protection. Nitrite can be applied successfully in open systems, but it requires very careful control and is usually selected only for special situations where other inhibitors cannot be used. Nitrite is relatively easily oxidized to nitrate in open systems and rendered ineffective. Also, nitrite is subject to biological degradation, which leads to both loss of inhibitor and biofouling problems.

Silicates are a class of molecularly dehydrated, polymeric forms of silica (SiO_2), similar in some ways to the polyphosphates described above. Silicates are weak cathodic corrosion inhibitors for carbon steel. They are sometimes used in low heat-transfer, once-through systems, but rarely in industrial open recirculating cooling systems. Silicates form a deposited film of insoluble silica that provides some protection but is not bonded to the metal and is easily damaged. This film forms slowly, so that in a corrosive system, rapid generation of iron oxide corrosion products can prevent effective silicate film formation.

Silicates are generally considered to be non-toxic materials. The forms of silica often found in makeup water are not useful for corrosion protection, but if present at sufficiently high levels in the circulating water, this silica can cause serious scaling problems (see below).

Other Corrosion Inhibitors For Carbon Steel

In addition to the individual components and synergistic mixtures discussed above, many other compounds have been used as carbon steel corrosion inhibitors. These are mostly organic materials, including carboxylic acid polymers, other polymers and non-polymeric compounds. It is beyond the scope of this chapter to discuss these compounds individually. Most of them are weak cathodic corrosion inhibitors, but when combined with other inhibitors they can produce pronounced synergism and very good, cost-effective corrosion protection. These organic products represent an important addition to available carbon steel corrosion control technology, especially in modern, alkaline cooling systems. Many of these products are protected by patents.

Corrosion Inhibitors For Copper and Copper Alloys

Inhibition of copper and copper alloy corrosion in open recirculating cooling systems is important for two reasons. First, corrosion of copper and copper bearing metals can lead to failure of heat exchanger tubes and other copper or copper alloy equipment in the system. Second, copper ions which go into solution as a result of this corrosion may galvanically deposit as copper metal on steel surfaces. This can create galvanic couples resulting in high rates of localized, or pitting corrosion of carbon steel. Good inhibition of copper and copper alloy corrosion is therefore an important factor in the control of carbon steel corrosion.

The copper corrosion inhibitors most often used in open recirculating cooling systems are mercaptobenzothiazole (MBT), benzotriazole (BZT), tolyltriazole (TT) and butylbenzotriazole (b-BZT). Some very effective proprietary copper corrosion inhibitors are also available. All of these products are organic compounds that form stable, adherent protective films on copper and copper alloy surfaces. These compounds can also form complexes with dissolved copper ions, thus inhibiting the deposition of copper on steel surfaces.

Selection of a specific corrosion inhibitor must be based on compatibility of the inhibitor with other chemicals in the system, operating conditions in the system and formulation considerations in preparing the complete inhibitor package.

Chromate-zinc combinations are also excellent corrosion inhibitors for copper and copper alloys, but as noted above have been eliminated from use for environmental reasons by regulatory agencies in many parts of the world.

Corrosion Inhibitors For Stainless Steels

Most cooling systems contain various grades of stainless steels in accessory equipment, and stainless steels are frequently used as heat exchanger tubing. As a rule, stainless steels are very resistant to general corrosion in an oxygenated water environment. Most stainless steels readily form a thin, protective oxide coating that provides good corrosion protection even under aggressive water conditions. For example, some of the more commonly employed chromium-nickel stainless steels may tolerate cooling water at pH values of 4 or less, as long as dissolved oxygen is present in the water. Some of the good carbon steel corrosion inhibitors may also provide additional corrosion protection for stainless steels.

While stainless steels as a group are relatively immune to general attack in oxygenated water, many are subject to rapid localized corrosion, or pitting, in an oxygen-deficient environment. Deposits of water-borne suspended solids, corrosion products from carbon steel corrosion and microbiological fouling can all create differential aeration cells leading to under-deposit corrosion of stainless steels.

For these reasons, the deposit inhibition and microbiological control portions of the cooling water treatment program are especially important for preventing localized corrosion of stainless steels. The potential for pitting corrosion of stainless steels increases with increasing dissolved solids and corresponding conductivity in the water, and with increasing temperature.

Corrosion Inhibitors For Other Alloys

Metals and alloys other than those discussed above are sometimes used in open recirculating cooling systems. While these alloys are not specifically discussed here, it should not be assumed that they are unimportant. All metals and other materials of construction used in the cooling system must be considered. Often, the less commonly used metals, such as titanium, do not require special corrosion control. In other cases, a specific metal: e.g., aluminum alloy tubing, or galvanized steel used in the cooling tower itself, may be the primary factor determining the selection of the optimum cooling water chemical treatment program.

Microbiologically Induced Corrosion

No discussion of corrosion control in open recirculating cooling water systems would be complete without mention of microbiologically induced corrosion (MIC). MIC is recognized as a major source of under-deposit pitting attack in many systems that are otherwise well protected against corrosion. All of the metals and alloys discussed above are subject to MIC, and many corrosion failures in alloys previously thought to be corrosion-resistant in open cooling water environments are now recognized as MIC-related.

MIC does not, in general, involve direct attack of bacteria on metal. Rather, MIC refers to corrosion that is induced or accelerated by the presence of products of microbiological metabolism. The most commonly seen cases of MIC are caused by sulfate-reducing bacteria. These are anaerobic bacteria that cannot live in the presence of dissolved oxygen in cooling water. They exist under deposits of corrosion products, suspended solids or biological slimes. They obtain their metabolic energy by reducing sulfate ions in the water and forming hydrogen sulfide (H_2S) or metal sulfide salts. The corrosion process generates more deposits and the process accelerates. Since the process is anaerobic (oxygen-free), corrosion resistant films that depend on dissolved oxygen in the cooling water break down. This leads, eventually, to deep pitting attack.

Other bacteria can also cause MIC to occur. Acid-producing bacteria, also anaerobic in nature, form organic acids under-deposits. These acids can attack carbon steel and other metals and alloys, producing the characteristic pitting attack. Iron-oxidizing bacteria react with ferrous (reduced) iron in cooling water and form voluminous deposits of mixed iron oxides and biological slimes, under which sulfate-reducing and acid-producing bacteria can grow.

Detection and recognition of MIC in operating cooling systems involves the use of several different techniques. Common microbiological assays of planktonic (free-swimming) bacteria in the cooling water are not a useful way to detect the presence of MIC-causing bacteria. A cooling system can show very low levels of planktonic bacteria and still be severely contaminated with anaerobic sulfate-reducers and other bacteria living in and under deposits. It is essential to use spool pieces and other test devices that can allow sessile (attached) bacteria to grow and to inspect the cooling tower, heat exchangers and low-flow points in the system regularly for the presence of biological deposits. Field test kits are available that can detect the presence of sulfate-reducing bacteria in fresh deposits removed from operating cooling systems, and in some cases, a metallographic examination can show corrosion patterns characteristic of certain types of MIC-causing bacteria.

The first line of defense in protecting cooling systems from MIC must be to keep the system clean and free of deposits. With the system clean, a microbiocide program must be selected which is effective under the operating conditions in the system and compatible with the water chemistry and with other treatment programs in use. Finally, the system must be monitored regularly to detect any appearance of biological or other deposits that can encourage anaerobic bacteria to grow. Microbiological control methods are discussed in detail in another part of this chapter.

Scale Inhibition

Five general methods are available for controlling mineral scale formation in open recirculating cooling systems. These are:

- External pretreatment of the makeup water.
- External sidestream or full flow treatment of the circulating water.
- Blowdown control.
- Chemical treatment of the circulating water to reduce the reactivity of one or more reacting species.
- Chemical treatment of the circulating water to "stabilize" reactive species so that they will not precipitate from solution.

Each of these methods will be discussed separately. The optimum scale control program developed for any specific system must depend on the makeup water composition and its availability, operating parameters in the cooling system, the number of concentration cycles to be carried in the circulating water, and sometimes on effluent considerations. Some systems, for which plenty of very soft makeup water is available, do not require any scale control program. For other systems, simple stabilization by chemical treatment (Item 5. above) is sufficient. On the other hand, in many parts of the country the makeup water is both hard and in short supply, so that it must be conserved. The most cost-effective scale control program for recirculating cooling systems in such cases may include, for example, partial softening of the makeup water (Item 1.), strict control of concentration cycles (Item 3.), sidestream filtration or softening (Item 2.) and stabilization chemical treatment of the circulating water (Item 5.).

- External Pretreatment of the Makeup Water

  External pretreatment was discussed previously. The most common external treatment of makeup water for open cooling systems is cold lime softening to remove hardness and alkalinity, thus reducing the calcium carbonate scale potential. The softened water can be used directly or blended with raw water to produce a desired level of calcium in the cooling tower makeup. In some plants, zeolite softened water is available, and this, also, can be used directly or blended with raw water. The use of very low or zero hardness makeup is generally not desirable. While this eliminates the calcium carbonate and calcium phosphate scale potential, it tends to make the water more corrosive and more difficult to treat, because many carbon steel corrosion inhibitors need calcium for good performance (see above).

  Reclaimed sewage effluent is finding increasing use as cooling tower makeup water. This water usually contains inorganic phosphates that can lead to calcium phosphate scaling problems in the cooling system. To avoid this problem, precipitation softening of the makeup water is used to remove phosphate before it enters the cooling tower.

- External Treatment of the Circulating Water

  Some cooling systems are equipped with precipitation softeners operating on a sidestream from the recirculating water, or occasionally on the full flow of recirculating water. Depending on how this equipment is operated, these softeners can be used to remove calcium, magnesium, alkalinity, phosphate, silica and suspended solids from the circulating water. The objective, in most cases, is to minimize water use and protect the environment while operating the cooling system at maximum cycles of concentration, with minimum or zero blowdown. This may require removal of silica and/or magnesium to avoid silica and magnesium silicate scale formation.

- Blowdown Control

  Increasing the blowdown rate from a recirculating cooling system is a simple way to reduce the levels of calcium and alkalinity in the water, thus reducing the calcium carbonate scaling potential. However, this is frequently not a cost-effective option. Increased blowdown, which means operating the cooling system at lower cycles of concentration, requires increased makeup water and produces more wastewater for disposal (see Figures 2 and 3). Increased blowdown leads to increased corrosion inhibitor usage.

  Blowdown control is, however, a critical part of any good scale control program in open recirculating cooling systems. It is important to strike a technically practical and cost-effective balance between the hardness that can be removed from the makeup water by pretreatment, the cycles of concentration that can normally be achieved, the amount and quality of blowdown water that can be tolerated and the costs of acid and stabilizing treatment chemicals. The ability of the plant to control the system is also an important factor. Widely varying blowdown rates can make any scale control program costly and/or ineffective. Also, lapses in feed of stabilizing chemicals can lead to serious scaling problems if the system is operating under supersaturated conditions.

- Chemical Treatment To Reduce Reactivity of Scaling Species

  Control of the pH of the recirculating cooling water with mineral acid is a simple and cost-effective way to reduce the scaling potential for calcium carbonate and calcium phosphate. The solubility of these two minerals increases with decreasing pH and alkalinity. The addition of mineral acids to cooling water removes an equivalent amount of alkalinity by neutralization. Calcium sulfate scaling cannot be effectively controlled with acid because the solubility of this salt is almost independent of pH.

  Acid, usually sulfuric acid, can be a viable part of a complete scale control program. However, reducing the pH increases the corrosivity of the water and may interfere with corrosion protection by making calcium/corrosion inhibitor salts too soluble. Also, sulfuric acid is a hazardous material and difficult to handle, especially in small systems. Other acids, such as sulfamic acid, can be used. These are safer to handle but more expensive than sulfuric acid. As a general rule, the most cost-effective and reliable combination scale and corrosion control programs for modern open recirculating cooling systems are designed to operate well into the alkaline range, using sulfuric acid only to trim the pH as needed. Another advantage of this method of operation is that if an acid spill should occur into the recirculating water, the buffering capacity of the water can usually neutralize considerable acid before the pH falls dangerously low.

  The scale-forming potential of ions such as calcium and magnesium can be reduced or eliminated by reacting these ions with complexing or chelating agents such as ethylenediaminetetraacetic acid (EDTA). EDTA forms very soluble, stable chelates with calcium, magnesium and other metal ions, thus making these ions effectively unavailable for other reactions. However, because EDTA is required in stoichiometric proportion to the ions to be chelated, this approach is not economically feasible and is rarely used in cooling water treatment.

- Threshold or Stabilization Treatment For Scale Control

  Threshold inhibition of mineral scale formation in open recirculating cooling systems is based on the use of substoichiometric amounts of compounds that impede either the formation or the growth of scale crystals. The concept, and the term "threshold treatment", are more than forty years old, but the exact mechanisms involved are still not fully understood.

  The maximum equilibrium concentration of any dissolved compound at a defined temperature and water composition is said to be its saturation concentration. Some compounds may exist for periods of time in solution at concentrations substantially above the equilibrium saturation concentration, even though these solutions are thermodynamically unstable. These solutions are said to be supersaturated. Threshold treatments, or threshold agents as they are often called, do not change the equilibrium solubility of sparingly soluble compounds. They simply allow supersaturated solutions of these compounds to remain stable for longer periods of time than they would without treatment.

  Threshold treatment requirements for any system can only be predicted by experiment and from empirical data. The amount of threshold agent required to achieve complete scale inhibition (the threshold concentration) varies with system conditions and increases as the degree of supersaturation of the scale-forming compound increases.

Many compounds used in water treatment are effective for more than one purpose. Some good threshold inhibitors are also good carbon steel corrosion inhibitors or good dispersants for water-borne suspended solids. When a compound shows both scale and corrosion inhibiting properties, the concentration required for corrosion inhibition is often many times higher than that required for scale inhibition.

Following is a brief discussion of some of the more common scaling compounds and corresponding threshold agents used in open recirculating cooling systems.

- Calcium Carbonate

  Calcium carbonate is by far the most common mineral scale found in open cooling systems. Fortunately, it is also the easiest to control. Acid addition shifts the equilibrium in equation 13 to the right and reduces the scaling potential. External treatment of the makeup and/or recirculating water can reduce both calcium and alkalinity levels. Finally, many threshold agents are available that provide excellent stabilization of calcium carbonate solutions. In many cases, threshold treatment alone is all that is required for calcium carbonate scale control.

  Following are some of the commonly used threshold agents for calcium carbonate stabilization:

  <u>Polyphosphates</u>

  Polyphosphates can effectively stabilize slightly supersaturated solutions of calcium carbonate. Relatively low polyphosphate concentrations are required, so that if polyphosphate is not also in use for carbon steel corrosion control, the amount of orthophosphate formed by reversion (hydrolysis) of the polyphosphate will not cause calcium phosphate scaling problems.

  <u>Organic Phosphorous Compounds</u>

  Many different phosphonates and phosphate esters are good threshold agents for calcium carbonate. The most often used phosphonates are AMP (aminotris[methylene]phosphonic acid), HEDP (1,1-hydroxyethylidene diphosphonic acid), and PBTC (phosphonobutane tricarboxylic acid). These phosphonates can provide complete stabilization of highly supersaturated calcium carbonate solutions.

  Phosphonates are relatively stable in alkaline open recirculating cooling water systems. AMP is attacked rapidly by oxidizing biocides and HEDP is more slowly attacked. PBTC appears to be quite stable to oxidizing biocides. Dosages of phosphonates must be carefully controlled to prevent precipitation and deposition of insoluble calcium phosphonate salts. The precipitation of the phosphonate can cause a loss of calcium carbonate control. Sometimes organic carboxylate polymers are added as threshold agents to prevent calcium phosphonate precipitation. This allows phosphonates to be used at higher calcium levels than would otherwise be practical.

  <u>Organic Polymers</u>

  Many organic polymers containing carboxyl and sulfonic acid groups show stabilizing properties for calcium carbonate. The performance of these products depends upon the type and arrangement of the functional groups, and on the molecular weight of the polymer. These polymers are not as effective as the phosphonates, but they seem to show synergistic performance and many phosphonate/polymer combinations are excellent threshold agents for calcium carbonate.

  Most of these polymers are stable in alkaline cooling water systems and they are not readily attacked by oxidizing biocides. As with the phosphonates, some polymers form insoluble calcium salts. Polymers and operating conditions must be selected properly to avoid this problem.

<u>Calcium Sulfate</u>

Calcium sulfate scale is a problem only in recirculating cooling systems carrying high levels of both calcium and sulfate ions. Threshold stabilization has been found to be the most cost-effective method for controlling calcium sulfate scale. Most of the threshold agents discussed above for calcium carbonate control are also effective on calcium sulfate, but their order of reactivity is different. The inorganic phosphates are not effective and phosphonates show only moderate activity. The most effective threshold agents for calcium sulfate scale are organic polymers containing carboxylic acid functional groups. Polyacrylic acid and modified polyacrylic acids are among the best polymers for this use; their reactivity varies somewhat with the molecular weight of the polymer.

<u>Calcium Phosphate</u>

Many surface water makeup supplies contain one or two ppm of orthophosphate. When cycled up in a cooling tower system the incoming orthophosphate may present a problem depending upon calcium levels and operating conditions in the system. It may create a potential for calcium phosphate deposition. Threshold agents are used to control this problem if it occurs. Higher levels of phosphate, such as from reclaimed sewage water used as makeup, are normally removed by external pretreatment using precipitation chemistry as explained above.

Equation 16 shows the equilibrium among the various phosphate species commonly found in solution in cooling water systems. Reducing the pH with acid shifts this equilibrium to the left and reduces the calcium phosphate scaling potential. For this reason, phosphate-based corrosion inhibitor programs are commonly operated with pH control. It is not, however, desirable to use pH levels below 7.0 because corrosion inhibition is less effective and polyphosphates tend to hydrolyze more rapidly in acid solutions. As a practical compromise, phosphate-based programs are often operated in the pH range 7.0 to above 8.0 pH units, with the addition of a calcium phosphate threshold inhibitor.

Organic phosphonates, such as AMP and HEDP, are weak calcium phosphate scale inhibitors. Carboxylate polymers, such as the polyacrylates and polymaleates, have some effect and can be used in neutral and slightly alkaline systems. The most important threshold agents for calcium phosphate scale are copolymers and terpolymers containing two or more functional groups. Many of these products are very effective. Their performance depends on the composition, molecular weight and geometrical arrangement of the monomers in the polymer molecule, and on operating conditions in the system.

Suspended Solids Control

Water-borne suspended solids, as defined here, includes silt and clay, corrosion products and other metal oxides, and insoluble process contaminants, but not biologically active solids. Treatments for controlling deposition of these materials are developed based on experience and empirical data, since the materials are heterogeneous and poorly defined. Physical characteristics, such as particle size and density, and surface properties, including electrical charge, are probably more significant than chemical composition. Progress is being made in understanding dispersion chemistry in aqueous systems and this is leading to the development of new and better dispersants. It does not seem likely, however, that dispersion chemistry will, in the near future, reach the level of technical sophistication now achieved for corrosion and scale control in cooling water systems.

Naturally occurring solids, including corrosion products, and process contaminants are discussed separately below, since control approaches are somewhat different.

<u>Dispersion of Naturally Occurring Solids</u>

The most common and cost-effective approach for controlling deposition of suspended silt, clay and metal oxides is to use polymeric dispersants. Most naturally occurring suspended solids contain negative surface charges. Low molecular weight anionic polymers adsorb on the surfaces of these particles and increase the net negative charge. This, in turn, increases the electrical repulsion between the particles and prevents them from agglomerating and settling. In this way, the suspended matter can be stabilized so that it can be removed from the system with blowdown and/or sidestream filters.

Natural polymers, such as tannins and lignins, have been used for years as dispersants for clay, silt and metal oxides in open recirculating cooling water systems. These products have now largely been replaced with more effective, although sometimes more costly synthetic organic anionic polymers. Polyphosphates (inorganic, anionic polymers) are also effective in this application and, as explained above, can serve multiple functions in the total cooling water program.

<u>Dispersion of Process Contaminants</u>

Insoluble process contaminants often consist of oily matter in addition to solids. Again, two approaches are possible for controlling deposition of these materials. The most common method is dispersion. Surface-active agents, often non-ionic in nature, are added to interact with charged agglomerates, causing them to be separated into smaller particles and dispersed or emulsified in the water. Anionic dispersants are also useful and cost-effective in these systems, especially when significant amounts of inorganic suspended solids are present. High velocity and turbulence enhance the dispersion process.

If the process contaminants contain large amounts of oil, organic polymers can be added to cause the oily matter to coalesce and separate from the water phase. Typically, part of the organic matter will float to the surface, and part, in combination with inorganic matter, will sink. Sidestream oil separators can be used to remove both the floating and settled material. Polymers for enhancing the oil separation are normally added to the separator influent, so that these reactions do not occur in the cooling water system itself. Cooling systems not equipped with oil separators should not use chemical treatments that can cause oil separation, since this could lead to organic fouling of equipment.

Microbiological Growth Control

Cooling water microbiocides are commonly classified as either oxidizing or non-oxidizing.

<u>Chlorine</u>

The most commonly employed oxidizing biocide is chlorine. Chlorine, in the elemental state, is a gas that rapidly dissolves in, and reacts with water, forming hydrochloric and hypochlorous acid.

$$C1_2 + H2O \text{ <----> } H^+ + C1^- + HOC1 \qquad (19)$$

The oxidizing power of the chlorine then resides in the hypochlorous acid. Hypochlorous acid is a weak acid and dissociates in a pH dependent equilibrium.

$$HOC1 \text{ <----> } H^+ + OC1^- \qquad (20)$$

The biocidal effectiveness of hypochlorous acid is greater than the effectiveness of the hypochlorite ion. As the cooling water pH increases, the degree of dissociation into hypochlorite increases. Therefore, the effectiveness of chlorine as a biocide decreases with increasing cooling water pH (see Figure10). This is not a major concern in recirculating cooling water systems using continuous chlorination due to the long contact times available. However, in once through cooling systems and in recirculating systems using slug chlorination, contact time is limited. In these systems, the more rapid kill rate of hypochlorous acid makes pH an important consideration.

Chlorine is a broad spectrum microbiocide. That is, it is effective for the control of bacteria, algae and fungi. While chlorine serves as a very cost-effective microbiocide in many cooling systems, there are several potential disadvantages that should be considered for any specific cooling system.

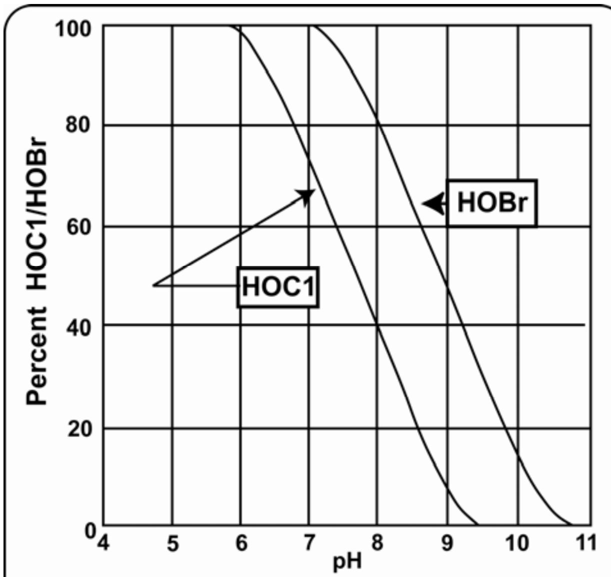

Figure 10
Ionization Of HOC1 And HOBr

- Chlorine is a very strong oxidizing agent and it may be consumed by compounds in the cooling water that are capable of being oxidized by chlorine.
- Chlorine may oxidize other cooling water treatment compounds.
- Chlorine will react with ammonia and amines to form chloramines. These compounds have weak biocidal properties and much of the biocidal effectiveness is lost when these reactions occur.
- Chlorine, if employed at excessive concentrations, will oxidize and damage cooling tower wood.
- Chlorine increases the corrosivity of water due to increased chloride concentration.
- Chlorine reacts with trace level organics to produce trihalomethanes that are EPA regulated carcinogens.
- Gaseous chlorine is extremely dangerous and good safety practices must be followed.

Chlorine may be purchased either as a compressed gas, solid calcium hypochlorite or sodium hypochlorite solution. The concentration of commercial grades of calcium hypochlorite and sodium hypochlorite are 70% and 12-15% available chlorine by weight, respectively.

Acceptable concentrations of residual chlorine discharge and allowable duration of discharge are regulated by the Environmental Protection Agency (EPA).

<u>Bromine</u>

Another frequently employed oxidizing microbiocide is bromine. Bromine, i.e., hypobromous acid (HOBr), can be generated by the reaction of chlorine with sodium bromide or released from solid bromine donor compounds as they dissolve. The primary advantage of bromine over chlorine is that bromine is a more effective biocide at elevated pH conditions. This is due to the fact that at high pH, hypobromous acid is much less dissociated to hypobromite ion (pK = 8.8) compared to hypochlorous acid (pK = 7.2) (see Figure 10). As indicated in the discussion on chlorine, halogen in the hypohalous form is the more biocidally active. Bromine compounds formed when ammonia and/or amines are present have greater biocidal effectiveness than the corresponding chlorine compounds. Bromine is not as strong an oxidizing agent as is chlorine. Consequently, its solutions are not as corrosive as chlorine. The tendency to form stable trihalomethane (THM's) is lower with bromine than chlorine.

<u>Chlorine Dioxide</u>

Chlorine dioxide is a gas that dissolves in water. Chlorine dioxide does not hydrolyze as does chlorine or bromine, and chlorine dioxide does not react with either ammonia or amines. Consequently, its microbiocidal effectiveness is not reduced either at elevated pH or in the presence of amines. Chlorine dioxide is not as strong an oxidizing agent as chlorine. As a result, the consumptive demand by reducing compounds in the cooling water is normally not as great as for chlorine. Chlorine dioxide does not form trihalomethanes to the extent that equivalent chlorine concentrations do. Chlorine dioxide, however, is very expensive, and is not stable. Chlorine dioxide must be generated on-site from sodium chlorite, a potentially dangerous oxidizing agent. In addition, chlorine dioxide is much more volatile than chlorine or bromine and can be rapidly depleted from recirculating cooling water by air-stripping.

<u>Non-oxidizing Microbiocides</u>

There are a wide variety of chemical compounds that are used as non-oxidizing microbiocides in open-recirculating cooling systems. Some of these compounds have broad spectrum biocidal effectiveness. Others are effective only for one or two classifications of microorganisms.

Consideration must be given to the compatibility of non-oxidizing microbiocides with the cooling water pH, other treatment products and potential system contaminants. Some non-oxidizing microbiocides react rapidly and may be used in systems with short half lives. Others require longer time periods and are applicable only in systems with relatively long half lives. Safety considerations in usage and discharge toxicity to aquatic organisms must be evaluated in the choice of non-oxidizing microbiocides.

Table VI is a tabulation of several classifications of non-oxidizing biocides employed in open recirculating cooling systems. The absence of an approved microbiocide from this list does not imply inferior performance or other disadvantages.

Occasionally, systems are treated by alternating treatment with two non-oxidizing microbiocides. This practice is based on the widely held, but not universal, belief that some microorganisms may develop immunity to a given microbiocide or that a more resistant population of microorganisms can be selected over a period of time. Alternate applications of two microbiocides is believed to minimize this potential.

A combination of oxidizing and non-oxidizing biocides can also provide cost-effective treatment approach for microbiologicalcontrol.

TABLE VI

CHARACTERISTICS OF COMMON NON-OXIDIZING BIOCIDES

Active	Primary Application	Effective pH Range	Comment
Carbamates	Bacteria and fungi	Above 7.0	Corrosive to copper.
Cocodiamine	Bacteria	6 - 9.0	Cationic charge.
Dibromoni-trilopro-pionamide (DBNPA)	Bacteria	6 - 8.5	Quick kill, hydrolyzes rapidly at high pH.
Isothiazolo-nes	Broad spectrum	6 - 9.5	Half-life 3-14 days, dangerous to handle.
Methylene-(bis)thiocya-nate (MBT)	Bacteria	6 - 7.5	Rapidly decomposes at pH > 7.5.
Quaternary ammonium salts (Quats)	Broad spectrum	7 - 9.5	Frequently foams, cationic charge, dispersive properties.
Tributyl tin oxide (TBTO) (TBTO)	Fungi and algae	7 - 9.5	Adsorbs on and protects cooling tower lumber, synergistic with quats -- combination broad spectrum.
Glutaralde-hyde	Broad spectrum	7 - 9.5	Partially inactivated by amines.

Surface active treatment components and/or dispersants are commonly employed in conjunction with both oxidizing and/non-oxidizing microbiocides. It has been demonstrated that these treatments may both enhance the performance of microbiocides by improving the penetration of deposits and contribute to better removal of deposits from systems surfaces.

Microbiocides and the identifying container label must be approved by the EPA for the specific application. That is, any biocide that is used in open recirculating cooling systems must be approved for this application and so identified on the container label. Acceptable use-concentrations are also specified on the container label. In addition to U.S. EPA registration, biocides still may require state EPA approval. Plant effluent streams that enter public water sources are regulated by the National Pollutant Discharge Elimination Systems (NPDES). Before a biocide is used, regulatory agencies should be contacted to determine if an NPDES permit is required.

Section VII.
CHEMICAL FEEDING AND CONTROL

Once an open recirculating cooling system is designed and the water treatment program is chosen, it is necessary to select an effective means of introducing the optimum amount of treatment chemicals into the cooling water. The importance of optimum treatment cannot be over emphasized. Too little treatment can result in a loss of heat transfer and destruction of heat exchange equipment. Too much treatment will result in waste of chemicals and may result in aggravation of the problems that the treatment is intended to solve.

A good chemical addition and control system can save many times its cost in chemicals, water, manpower, equipment life and production. However, no system can substitute for human observation and judgment. The system is another tool, and like any tool, it must be checked, updated and recalibrated at regular, frequent intervals.

Every chemical feeding and control system has seven primary and several secondary elements.

Sensor measures some property of the circulating water and indicates either that more or less treatment is needed, or that the level of treatment is satisfactory.

Measurement receives a signal from the sensor and converts this signal into a meaningful output signal.

Controller receives the output signal from the measurement device and directs the controlled element to operate. It may or may not report any change.

Data Logger/Logic Controller also receives the output signal from the measurement device. Data is recorded in a database for future reference. A logic circuit may be included that compares the data against known requirements and resets the controller to adjust to current data.

Controlled Element receives instructions from the controller and operates the feeder in accordance with the instructions. It may also operate a fail-safe system to cope with emergencies and, as a last resort, may shut down the system.

Feeder at the direction of the controlled element, introduces the chemicals to the diffuser. Usually this includes chemical storage.

Diffuser takes the chemicals from the feeder and distributes them in the circulating water.

These seven elements are shown schematically in Figure 11.

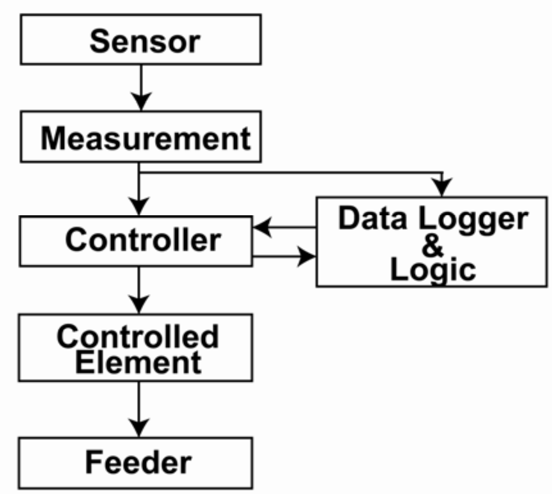

Figure 11
Seven Elements Of Chemical Feed System

The acid feed system (Figure 12) used in many cooling water operations, is a good example of a chemical feed control system.

<u>Sensors</u>

There are numerous parameters that may serve as the basis for control of open recirculating cooling water systems. These parameters may be classified as either physical, hydraulic or chemical. Some chemical feed programs may simply be based on addition of a specified amount of chemical at regular time intervals.

Physical, hydraulic type sensors include:

- <u>Flow</u> - In general, the chemical feed requirement is proportional to the makeup flow. Most chemical feed systems are designed to feed on this basis. Flow sensors include orifices venturis, flow tubes, turbine meters and meters based on acoustical measurements.

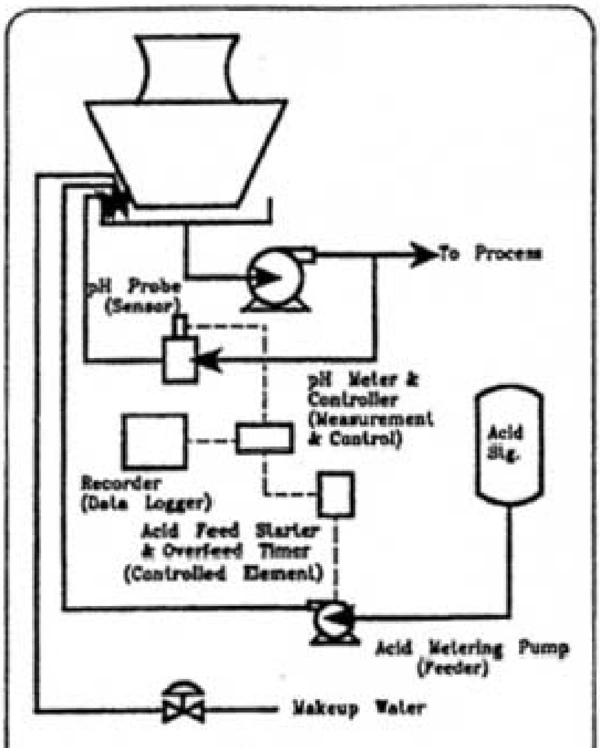

Figure 12
Acid Feed System

- Level - The makeup flow is frequently based on cooling tower basin level. Consequently chemical feed may be based on a float type level sensor in the tower basin.

Sensors which measure some chemical property of water include:

- Conductivity - The electrical conductivity of the water is a measure of the total ionic concentration of the water. Both corrosion rates and scale formation potential for any specific system, are normally proportional to the conductivity of the water. Conductivity sensors are the most frequently employed chemical detection device. This sensor is most often used to operate the blowdown system for control of cycles of concentration.

- pH - Carbon steel corrosion rates decrease with increasing pH, and the scale potential for most scale forming compounds increases with pH. Consequently, pH measurement is often the basis for addition of acid or caustic for chemical control of open recirculating cooling systems.

- Chlorine Residuals - Chlorine is the most commonly employed biocide for open recirculating cooling systems. Chlorine analyzers are commercially available that measure either free or total chlorine residual on a continuous basis.

- Turbidity - Turbidity measurements provide an indication of the amount of suspended matter in cooling water. Turbidity measurements, therefore, provide an indication of the deposit potential.

Specific Ions and Compounds:

- Numerous specific ion analyzers are commercially available to measure individual ion concentrations in water. Many analyzers are also available to measure either compounds or classes of compounds dissolved in water.

- Corrosion Rate - Corrosion rate sensors are available to provide continuous, instantaneous measurement of corrosion rates for any alloy. These sensors are normally based on linear polarization resistance between a pair of electrodes.

  The location of the sensor in the cooling system is very important. Sensors that measure treatment chemicals should be located at a point after the treatment is well mixed. Corrosion rates increase with increasing temperature. Therefore, corrosion monitors should be installed at the outlet of the highest temperature heat exchanger. Careful, individual consideration should be given to the need for measurement, the reliability of the sensor and the location for each sensor employed.

Measurement Devices

Signals from sensors may be electronic, hydraulic or mechanical in nature. Signals may also be the result of manual measurements and observations. Signals from continuous sensors are usually very weak. The first function of the measurement device, therefore, is to amplify the signal sufficiently to operate the measurement system. Sometimes the nature of the input signal is changed. For example, the input might be air pressure to operate a pneumatic valve. The second function of the measurement device is to provide an output signal to the controller and datalogger/logic devices.

- Controller - The function of a controller is to direct the addition of chemicals. The controller can be programmed to add chemicals continuously, proportional to water flow, in timed slugs, or as a combination of these methods. The controller may be programmed to add two or more chemicals alternately. In all cases, the purpose of the controller is to maintain control of critical operating parameters in the cooling water system by adding chemicals in response to signals that indicate the condition of the system.

- Datalogger/Logic - The function of the Datalogger/Logic is to report results. The device may be either an indicating or a recording instrument.

The objective is to report how well the desired control is being maintained. Control logic can be added to alter the level of treatment if the sensor results indicate that corrective action is required. In advanced computerized systems, the logic program can provide graphical techniques for analysis of system control.

Controlled Element

This element is usually either a valve or a pump regulating transfer of chemicals from the feeder to the diffuser. The valve opening or stem position may be regulated magnetically, pneumatically or hydraulically. It may be either motor driven or provided with a mechanical linkage. The pump must be suitable for handling the material being transferred. The quantity of chemical delivered may be controlled by varying the stroke, the speed, or the time of pump operation.

Another function of the controlled element should be the fail-safe feature. If the primary controlled element cannot keep operating parameters within the defined limits, the fail-safe system takes over. For example, an acid overfeed may cause the pH to drop below the allowable range. The fail-safe system will shut down the acid pump, open the blowdown valve and may introduce a neutralizing chemical.

Feeder

Chemicals may be solid, liquid or gaseous, requiring a variety of feeder types. Feeders can operate by gravity, eduction, by-passing, positive displacement or combinations of these methods. A gravity feeder may be simply a tank whose contents flow by gravity into the cooling tower basin. The rate of flow may be adjusted with a valve. The greatest difficulty with this simple system is that the rate of flow decreases as the liquid level in the tank falls. This difficulty may be overcome by adding a constant head system. A feed system employing a backpressure control is shown in Figure13.

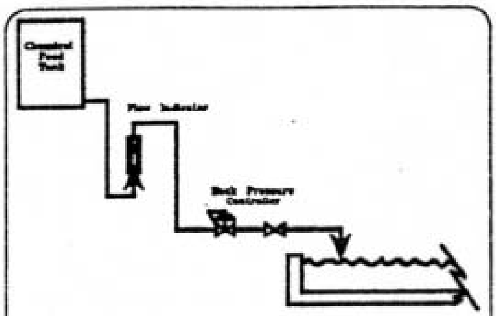

Figure 13
Gravity Feed System
With Constant Back Pressure

The eductor principle takes advantage of the fact that a low pressure area exists in a line of flowing water just after it passes a restriction. Liquid or gaseous chemicals may be drawn into this low pressure area and mixed with a water stream. Eductor systems are able to feed chemicals in proportion to water flow. This is the usual way of introducing chlorine into cooling water.

Bypass feeders take advantage of either the pressure drop across a restriction in a flowing line or the pressure difference between the inlet and outlet of a pump. A bypass feeder useful for slowly dissolving solids is shown in Figure 14. A bypass feeder for liquids is shown in Figure 15.

Positive displacement feeders are simply pumps. These are discussed in the section on controlled elements. A gravity feeder can be converted to a positive displacement feeder by substituting air pressure over the chemical tank as the motive force.

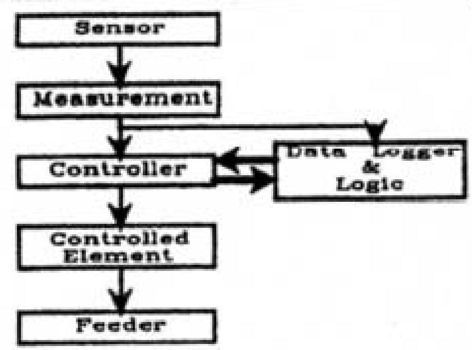

Figure 14
Seven Elements Of
Chemical Feed System

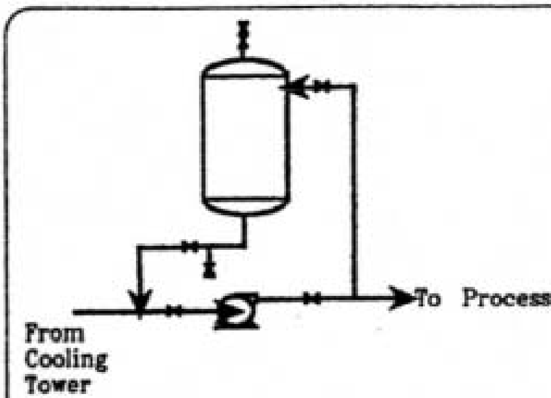

Figure 15
Bypass Feeder – Liquid

Diffusers

Simply adding chemicals to the circulating cooling water is often not sufficient to obtain the desired level of control. Chemicals should be added so that they are uniformly mixed with the entire stream of water. Educted chemicals are well dispersed, as are chemicals added to the inlet of a centrifugal pump. Perforated pipes, in various configurations, are useful for dispersing the chlorine solution from chlorinators in the basin.

The addition of sulfuric acid is quite difficult because of its high density and great corrosivity. One good method is to use a slightly inclined wooden trough, at least 10 feet long. A good flow of water, either makeup or circulating water, enters the upper end of the trough. The concentrated acid is also added near the upper end. The water and acid mix as they flow down the trough and are discharged into the cooling tower basin (Figure 16).

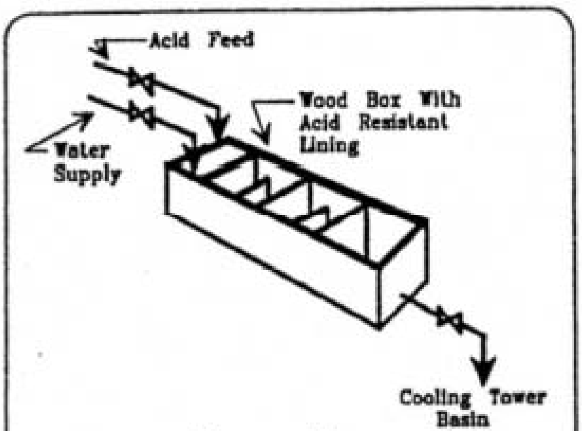

Figure 16
Acid Mixing Trough

The optimum point of chemical application depends on the potential problems, the most critical equipment and system design. Careful consideration should be given to choosing the feed point for each chemical so that maximum benefit is obtained in the critical equipment, with minimum loss through blowdown. The compatibility of the various treatment chemicals used must be considered when injection points and time of injection are being determined.

<u>Summary</u>

The success of the chemical treatment program is often dictated by the reliability and accuracy of the chemical feed system. Chemical feed systems can be as simple as operator sampling and testing for each control parameter, followed by manual addition of the proper chemical at the proper dosage, or it can involve full computer control for measurement and adjustment of all chemicals included in the treatment program. It is important to remember that operators are the most important element of the treatment program. They must have a complete understanding of the treatment program and how the use of each chemical affects the cooling water system performance. Operators must know how their chemical feed system operates in order to recognize whether or not it is working properly. Also, a chemical feed system needs periodic maintenance and recalibration. It must be the operators' responsibility to recognize when maintenance is needed.

Section VIII.
PERFORMANCE MONITORING

General Concepts

The ultimate measure of the performance of a cooling water chemical treatment program involves production records of the specific process units, operating cost records, and inspection records of heat exchangers and other critical equipment. On a long term basis, these data provide an on-going record of the success or failure of the system and chemical programs involved in specific process operations.

Short term, however, it is very important to be able to predict system performance on-line so that costs can be optimized and corrections can be made as needed to avoid failures and maintain good control of cooling water operations. Two general methods are available for collecting on-line performance information:

- Measurements of cooling system operating variables and water composition.
- Direct measurements of corrosion, scale formation, etc. using monitoring devices installed in the recirculating cooling water system.

To illustrate the chemical monitoring approach, it might be assumed that an acceptable chemical treatment program has been defined and implemented, and that if certain defined chemical parameters are consistently maintained, then acceptable results will be achieved. For example, a corrosion inhibition program might specify that the inhibitor concentration be maintained at 50-60 ppm. Monitoring for this system might consist of hourly or daily measurements of inhibitor concentration. It might be assumed, based on other knowledge, that if the inhibitor concentration is maintained within the control range, an acceptable corrosion rate of <2 mpy will be achieved.

At the same time, corrosion rates in the system can be monitored directly. In the example above, instantaneous corrosion rates might be continuously measured and the inhibitor concentration varied as required to achieve the desired 2 mpy corrosion rate.

In practice, a successful cooling water control program normally includes both chemical controls and performance monitoring. A typical program might require:

- Continuous or frequent measurement of chemical treatment pumping rate.
- Continuous or frequent chemical analyses of important chemical parameters.
- Measurement of corrosion rates.
- Observation of scaling and deposition rates, using deposit monitors.
- Regular counts of planktonic and sessile bacteria levels.

Chemical treatment programs are usually selected on the basis of plant process performance objectives plus predictions of the most cost effective program required to meet the defined objectives. These predictions may be made by the water treatment supplier, corporate personnel, or both, and may be based on plant or other local experience, published technical information or data submitted by the supplier. Thus, when the program is selected and implemented, monitoring and control methods must also be established and implemented to confirm the predicted performance.

More frequent testing, or additional daily tests, may be specified by the supplier. Tests to be run by the supplier might include, in addition to the daily tests, chloride, sulfate, iron, copper and other specific parameters.

Corrosion Measurement

The corrosivity of a cooling water to specific metals can be established by measuring loss in weight of exposed metal coupons or test heat exchanger tubes. Electrical measurements of the change in resistance or the polarization of exposed metal specimens can also be used as relative measures of corrosion rate.

Corrosivity by direct weight loss is determined either by the exposure of specifically prepared and weighed strip or bar coupons in a moving stream of water, or by the exposure of specially prepared and weighed tubes under heat transfer conditions in test heat exchangers. In either case, deposits on the coupons can also be studied to determine their relative rates of formation and composition. The type of corrosion can be determined by the condition of the surface after deposit removal. Weight loss measurements provide average corrosion rates for the metals involved during the exposure period.

Two different types of electrical measurements can be used to estimate corrosion rates in on-line equipment. First, the change in electrical resistance of a metal strip or wire exposed to a corrosive medium can be related to the change in thickness or diameter of the specimen, and hence to the corrosion rate. Probes and instrumentation based on this electrical resistance method are readily available. The data are interpreted as average corrosion rates for the period of exposure, similar to corrosion coupons. Electrical resistance probes can be used in cooling water systems, but they are more commonly selected for process side and high temperature applications.

The second electrical method for observing on-line corrosion rates involves passing a small current between two electrodes exposed in a conductive medium, i.e., cooling water. As films and deposits form on the electrode surface, the electrodes become "polarized", and current flow decreases. This is the "polarization resistance" method. From the change in the current, an "instantaneous" corrosion rate can be calculated, that is, the corrosion rate in effect at the moment of measurement rather than the average rate over the time the electrodes were installed.

In the following paragraphs, the essential steps in each test method are outlined and the variables that affect the accuracy and reproducibility of the methods are discussed.

When using any of these test methods, it must be remembered that the measured corrosion rates are only for the coupons or electrodes themselves under the existing conditions. These rates may or may not be characteristic of other locations in the system. For example, since coupons are not heated, they may not reflect the corrosion rate in the heat exchanger. Therefore, it is frequently found helpful in evaluating cooling water treatment performance to use two or more test methods and to expose test coupons under several different sets of conditions.

The discharge water line leaving the most critical heat exchanger or condenser will provide the most severe conditions. If this point is not accessible, the cooling tower return water header should be considered. Monitoring at the discharge of the recirculating pump generally includes less severe conditions since this point often sees the lowest temperatures and usually the highest chemical residuals in the recirculating cooling water circuit.

<u>Coupon Methods</u>

Standard procedures for measuring the corrosivity of cooling water by the coupon test method are outlined in the <u>ASTM Annual Book of Standards</u>, Volume Nos. 11.01 and 03.02 as method D2688, "Standard Test Methods for Corrosivity of Water in the Absence of Heat Transfer (Weight Loss Methods)". "Recommended Practice for Conducting Plant Corrosion Tests", No. A224-46 may be found in ASTM Standards, Part 3, Page 173 (1966). A "Procedure for Measuring Corrosivity of Cooling Waters by Use of Metal Specimens" is given in the NACE Cooling Water Treatment Manual.

Installation

- Bypass Type Assemblies
  Test coupons may be exposed in several types of assemblies. The most widely used is the bypass type manufactured of PVC pipe, nipples, tees and elbows as shown in Figure 17. These racks can be assembled to expose one or more coupons and are generally located where the water can be returned to the system. A convenient location is the cooling tower hot water return line. When 1 inch pipe is used for the assembly, a flow of 8 gpm results in a velocity past the coupon of about 3 feet per second. The velocity in the bypass assembly should be comparable to the velocity in the system being monitored. A velocity of 3 feet per second is considered a mid range velocity. Figure 17 also shows the coupon holder, which is usually made of 3/8 inch to 1/2 inch by 6 inches long Micarta, or hard plastic rod, screwed into a suitable pipe plug. The other end of the rod is filed flat and then drilled and tapped for a 1/4 inch plastic machine screw which holds the coupon in place. When the coupons are in place, installed in horizontal pipe, the coupon face should be vertical to offer the least amount of available surface for deposition.

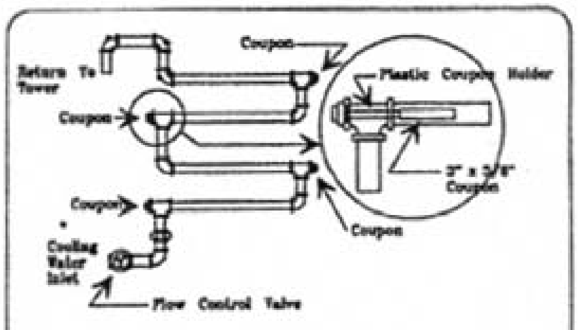

Figure 17
Corroson Coupon Rack

The use of carbon steel for coupon rack construction offers the additional advantage that sections of rack can be constructed as removable "spool pieces" that can be removed periodically for inspection, photographed and returned to service.

If a bypass assembly is installed on the discharge of a critical heat exchanger, care should be taken in materials of construction. PVC is recommended for service below 150°F and 150 psi. If temperature or pressure is greater than this limit, the bypass assembly can be constructed of mild steel pipe using Micarta or hard plastic rods for coupon attachment. The bypass assembly can be installed across a valve or booster pump. Care should be taken in controlling the flow through the coupon assembly. Flow should be regulated by a rotameter or other visual flow control device. One coupon exposure point can be reserved in the rack for installation of an instantaneous corrosion probe. Probe readings can be helpful, not only in long term monitoring, but also in the detection of upset conditions between the dates of coupon removal. In most cases, coupons are used to measure average corrosion rates, while the probes are used as an "early warning device" to detect sudden changes in system conditions, i.e. acid spills, that could affect corrosion rates if allowed to continue.

- Retractable Probe Assemblies
  A second method used for the exposure of corrosion coupons directly in water lines is the retractable probe assembly shown in Figure 18. The assembly is normally composed of a stainless steel rod, threaded for a hard plastic or nylon coupon coupling, and a packing gland fitted with a safety Swagelok fitting. The packing gland body is large enough to admit the coupon when it is withdrawn from the pipe through the shut-off valve. The packing gland is screwed into the receiving gate valve attached to the pipe wall.

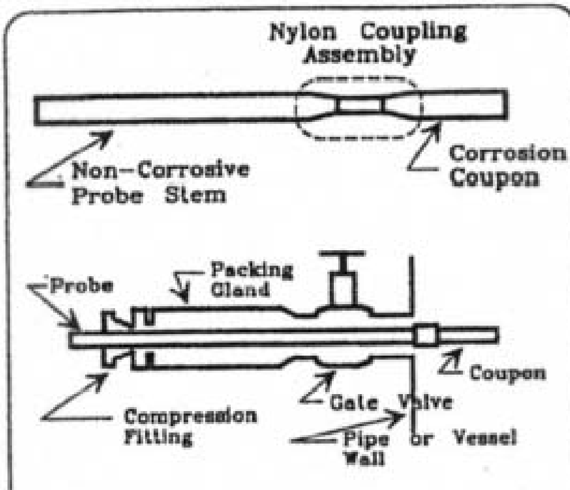

Figure 18
Retractable Coupon And Probe Assembly

Fabrication and Preparation of Corrosion Coupons

Flat coupons are normally cut from sheet metal or cold rolled flat wire. Sizes are normally 3/8 inch to 1/2 inch wide by 3 inches to 4 inches long and 1/16 inch to 1/8 inch thick. Holes for attaching the coupons to the holders are normally 1/4 inch in diameter. ASTM method D2688 provides standard directions for preparing corrosion coupons.

Square or round coupons are made of cold rolled rod or bar stock and may be 1/4 inch to 1/2 inch by 3 inches to 4 inches long. They are threaded on one end for convenient mounting into plastic holders.

Coupons of the chosen size, which have been pre-weighed to the nearest 0.1 milligram and have had the surfaces prepared and cleaned according to the techniques recommended by the National Association of Corrosion Engineers, are supplied by qualified water treatment companies or may be purchased from various metal specimen manufacturers. Ferrous coupons are supplied in envelopes made from vapor phase inhibitor-impregnated paper and non-ferrous coupons are supplied in a sealed container ready to use.

Most common heat exchanger metallurgies can be duplicated with corrosion coupons. To assure comparative results, once a coupon type, size and source has been chosen, it should be continued for a given series of tests.

Coupon Exposure

Coupons should not be touched directly during the mounting and installation process. When coupons are kept in small plastic or treated paper bags, the bags can be cut off at the hole end of the coupon and the bag used to protect the coupons during attachment to the holder rod.

- The sequence for installing the retractable probe is as follows:
- Draw the probe back until the coupon is inside the packing gland housing.
- Screw the packing gland into the gate valve and open the valve.
- Push the probe in until the coupon and coupling are inside the pipe wall and then tighten the nut of the Swagelok fitting to hold the probe in place.

The minimum duration of exposure for any one coupon should be twenty-eight days. If two or more coupons are run for the same length of time in a bypass assembly, the coupon over which the water passes first may show the highest corrosion rate because of the drop in temperature as the water passes through the assembly. Frequently, when running long term tests, four coupons are changed in a sequence of four, eight, sixteen, and thirty-two week intervals. Normally, the longer the exposure period, the lower the measured average corrosion rate and the more closely the coupon will approximate conditions in the system.

It is critical that the flow rate past the coupons remain constant, and that flows be comparable to the flows encountered in the area of the cooling system being monitored. Low velocity produces less inhibitor contact on the surface, and false high corrosion rates may result. High velocity results in more turbulent flow, and the inhibitor or protective film may be stripped from the coupon surface, again leading to false high corrosion rates.

Great care must be taken not to loosen deposits from the coupon surfaces during the removal process. The exposed coupons should be placed in sealed containers and returned to the laboratory or the water treatment supplier, where they are cleaned, dried, weighed and inspected according to procedures given in ASTM D2688.

Visual Examination of Coupons

The type of corrosion attack on the surface of exposed coupons should be observed and reported as outlined below. Alternative terminology according to ASTM recommendations is illustrated in the attached Figure 19. It should be noted that several types of corrosion may occur on the same coupon.

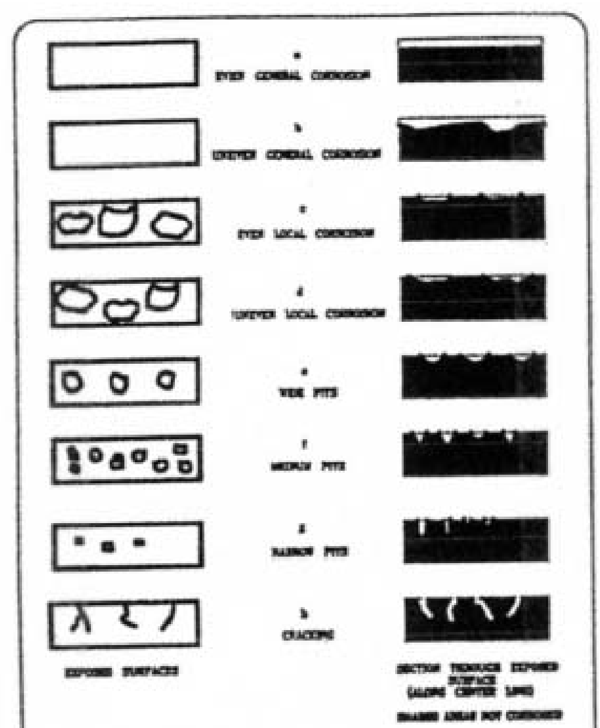

Figure 19
Types of Corrosion Attack

- General etching leaves the coupon surface smooth and uniform, with no special features other than a generally etched appearance.
- Pitting attack is described by its maximum depth, average width, and the number of pits per square inch.
- The depth of the pits is determined mechanically or by optical methods. A convenient mechanical device is an Ames, No. 262 calibrated dial depth gauge or a

micrometer depth gauge (cf. Materials Performance, Volume 14, Page 16 [November, 1975]).

- Local attack results in lake-like depressions that are frequently the result of tuberculation. The width and number of these depressions per square inch are reported.
- General local attack occurs when local attack or wide shallow pitting appears to cover the entire surface of the coupon.
- Under-deposit attack occurs when foreign material or corrosion products adhere to the coupon surface and corrosion attacks the area covered.
- End attack and edge attack are frequently observed on corrosion coupons, because these locations are stressed and therefore are more active than the general surface. The attack is similar to that found under acid conditions, where the edge of the metal is roughened by small, deep, sharp pits.
- Bacterial attack usually occurs under colonies of sulfate-reducing or acid-producing bacteria. The attacked area appears to made up of a series of concentric rings. The attack may be severe enough to cause penetration of the coupon.
- Crevice corrosion appears under or at the edge of the coupon, or in the contact area between the coupon and the holder. Crevice corrosion is important and should always be reported when it is observed.

Calculations

Corrosion rates are normally calculated from a corrected weight loss that takes into consideration the weight loss of a new coupon that is subjected to the same cleaning procedure as the coupon used for monitoring. Rates are reported as an average penetration in mils per year (mpy) or milligrams per square decimeter per day (mdd), assuming that localized attack or pitting is not present and that the corrosion is general.

Reporting Results

The corrosion coupon report should include the following:

- Company and Plant location.
- Identity of cooling system.
- Location within system.
- Coupon number.
- Type metal.
- Dates of exposure.
- Length of exposure.
- Temperature of water.
- Water flow rate.
- Type and color of deposits.
- Average corrosion rate, mpy and/or mdd.
- Average pitting rate, mpy and/or mdd.
- Visual appearance of cleaned coupon. Type and severity of attack.

Electrical Methods

A number of commercial instruments are available which give instantaneous corrosion rates (in mils per year) usually by measuring the potential difference between test electrodes and reference electrodes which meet the requirements of ASTM D2776 for Corrosivity of Water in the Absence of Heat Transfer (electrical methods). The manufacturers of these instruments state, that under certain conditions and with experience, readings may also be used to indicate the pitting tendency of the circulating water. These instruments may be installed with recorders and/or alarms using the corrosion rate to monitor treatment continuity as well as efficiency. They may also be equipped to control treatment dosage to maintain a specific corrosion rate and thus maximize treatment efficiency.

Since the corrosion rate indicated by these instruments is an instantaneous corrosion rate, its use should normally be supplemented by a conventional coupon installation to get a long term average corrosion rate. Instantaneous corrosion rates and average rates from coupons do not always agree closely, because of differences in the detection methods. As the instantaneous probe electrodes "age" in the cooling water, the measured rates tend to approach long term coupon corrosion rates.

In addition to corrosion rates, instantaneous corrosion rate instruments provide valuable early warning of changes in operating conditions. For example, a sudden rise in the measured instantaneous rate is a clear signal that a problem exists. Possible causes might be, for example, loss of inhibitor feed or an overfeed of acid. Some corrosion meters can be wired to trigger alarms if the corrosion rates rises above a preset value.

Fouling Evaluations

Corrosion and deposition are closely related in most cooling systems. Dirty systems, as a rule, show higher corrosion rates and more localized corrosion than clean systems.

On-line, sidestream heat exchangers are often used to evaluate deposition rates in cooling water systems. A plot of heat transfer rate versus time can indicate the extent of deposition in an on-line heat exchanger, but cannot indicate the nature of the deposits. An on-line test heat exchanger is a valuable tool in assessing deposition tendencies on heat exchange surfaces in cooling systems. Both visual inspection and analysis of any deposits formed can assist in identifying and controlling deposition in the operating system.

Hardness salts, primarily calcium carbonate, exhibit inverse solubility; that is, they are less soluble at high temperatures than at lower temperatures. Scale formation occurs, therefore, on the hottest portions of a system. Operation of a monitoring unit at higher temperatures or higher heat fluxes than actual plant conditions allows deposition tendencies to be accelerated. However, accelerated testing must be used with great care. Severe conditions may exaggerate problems and eliminate solutions that would be acceptable under realistic conditions.

Several systems have been devised to measure both corrosion and fouling on heat exchange surfaces. Most of these systems are simply small heat exchangers with steam or electrical energy as the heat source. Visual inspection of the on-line unit can be accomplished by placing the heat exchange surface inside a glass or Plexiglass tube with water flowing through the annular space.

The variables in the system (cooling water flow rate, heat flux and system metallurgy) can be set to establish desired test conditions. This type of monitoring device can allow fairly rapid determination of fouling tendencies. Changes in the skin temperature of the heat transfer surface can indicate the extent of fouling or deposition as it occurs. The deposits that form on the tube surface can be analyzed. In this way, both the extent and nature of the deposition can be evaluated.

Microbiological Monitoring

Microbiological monitoring in cooling systems is generally conducted either by monitoring microorganism counts in the bulk water or by directly measuring biofouling using a biological deposit monitor. Monitoring bulk water organism counts is the traditional and most widely used approach and is generally conducted using the standard plate count (SPC) method. Other biocount assay methods, including the popular dip stick method, are also used (Table VII) with results of the analyses usually expressed as CFU (colony forming units) per ml. Biocount results are used for establishing microbiological trends for a system, for quantifying and differentiating between the different types of troublesome organisms, and for evaluating biocide effectiveness. Experience with a particular cooling system will indicate what level of bulk water organisms should be maintained and not exceeded to ensure acceptable biocontrol for that system.

Legionellae bacteria in cooling water systems can also be monitored using plate count methods by utilizing special culture media selective for the growth of Legionella. An alternate approach is a microscopic technique called the direct fluorescent antibody (DFA) test. Either approach, when properly performed, permits enumeration of *legionellae* in order to determine whether biocide treatments are effectively keeping the numbers of *legionellae* in check.

The second approach to microbiological monitoring in cooling water systems is to directly measure microbiological control using a biodeposit or biofouling monitor. Several different types of biofouling monitoring devices are used and are based on monitoring either changes in pressure drop or heat transfer resistance, or direct measurement of the biofilm using surface colonization techniques (Table VIII). The biofilm monitor operates on a slip stream from the cooling system and can provide real time assessment on system biofouling conditions. Biocontrol treatment program performance can then be monitored and adjusted to maintain clean surface conditions in the unit. In combination with an appropriate bulk water monitoring program, complete cooling system microbiological status can always be known and treatment program fined-tuned for optimum performance.

Finally, special methods have been developed to detect the presence of sulfate-reducing bacteria (SRB), acid-producing bacteria (ACB) and other anaerobic species associated with microbiologically influenced corrosion (MIC). These methods all require careful in-field sampling techniques. The methods involve the use of either special growth media or enzymatic reactions that generate a gas or a color change that can be measured quantitatively.

TABLE VII
COOLING WATER
MICROORGANISM ENUMERATION METHODS

METHOD	METHOD BASIS	ADVANTAGES/DISADVANTAGES
Standard Plate Count, Total Plate Count, Agar Tube Count	Organism growth on culture media	Accurate, widely used, selective growth media can be used for differential analysis / requires dilutions, cumbersome for field use, 48-72 hours incubation required for results
Dip - Slides	Organism growth on culture media	Convenient for field use, widely used / 48-72 hours incubation required for results, not as accurate as plate or agar tubes
Metabolic Quick Tests[a]	Detection of specific cellular components, enzymes, or metabolic end-products	Results available in as little as 15-20 minutes, many tests species specific / may require expensive equipment, lower limit of detection may be high

[a] Examples include ATP photometry, respirometry, fluorescent antibody tests, and redox indicator tests.

TABLE VIII

COOLING SYSTEM BIOFOULING MONITORS

TYPE	PARAMETER MEASURED	ADVANTAGES/DISADVANTAGES
Differential Pressure[1,2]	Changes in fluid frictional resistance	Extremely sensitive to biofouling, simple and easy to use, real time fouling assessment, can provide electronic signal for data acquisition / no information on types of fouling organisms
Heat Transfer[3]	Changes in heat transfer resistance	Based on important cooling system parameter, real time assessment, can provide electronic signal / due to heated surface scale formation possible, no information on types of fouling organisms
Robbins Device[4]	Removable coupons for direct measurement of biodeposit	Contains test section with removable colonization coupons which can be used for deposit analysis, microorganism enumeration, etc., coupons can be used for information on microbial corrosion / coupon analysis required

[1,2] See: IX. References and Bibliography - H. Biofouling Monitoring, No.s 1 & 2

[3] See: IX. References and Bibliography - H. Biofouling Monitoring, No. 3

[4] See: IX. References and Bibliography - H. Biofouling Monitoring, No. 4

Chemical Analysis

Representative sampling and accurate chemical testing are essential parts of a sound cooling water treatment program. Each tower should be tested regularly to ensure that the proper chemical levels are being maintained and that the pH, alkalinity, conductivity, and other parameters of the water are within recommended ranges. The sampling and testing frequencies suggested in this section are only guides and should be adjusted as necessary to maintain an effective water treatment program.

Chemical analysis programs must be defined as required for each system. In large industrial cooling systems, analysis for selected control parameters are normally performed on a daily or shift basis by operating personnel. HVAC systems generally may be tested less frequently. More extensive analysis should be performed on a regular schedule by the water treatment supplier or by the plant or corporate laboratory. The makeup water should also be tested regularly, especially if seasonal variations affect makeup water quality.

When tests show that operating conditions are outside the established limits, corrective action should be taken. The unacceptable level should be brought back into the established limit. The results of all tests should be recorded to provide supervisory personnel with evidence that the water has been treated according to plant recommendations.

Most of the required analytical tests can be performed at the cooling tower site. Some tests, such as pH and conductivity, must be performed at the site immediately after sampling. The others can be performed at a central laboratory.

Typical daily tests in open recirculating cooling systems should include P and M alkalinity, calcium and total hardness, conductivity and inhibitor level. Other common daily tests include turbidity, COD (to check for process leaks), silica, chlorine residuals, etc.

pH testing is a special case. In large industrial towers, the pH should be recorded continuously and checked manually once each shift to confirm the recorded data. If acid or caustic is fed for pH control, even more frequent manual pH checks may be advisable to prevent a sudden loss of pH control.

Tests to be run by the supplier might include, in addition to the daily tests, chloride, sulfate, iron, copper and other specific parameters.

Periodically, samples should be cross-checked for accuracy by an outside laboratory. Some vendors provide such cross-checking as a service, and some laboratories participate in regular round robin testing programs.

All analytical data should be recorded on log sheets and plotted versus time. Computer programs are readily available to facilitate the graphing process. Data trending, as this process is called, greatly increases the value of the analytical data by highlighting problems and needs for action to improve control of the cooling water system.

The person doing the testing should be provided with separate laboratory space to perform the routine control tests that are required. The space should preferably be set apart from the plant or the control room by walls. It should have ventilation and comfort heating and cooling, so the tests can be performed under controlled conditions. Lights should be provided at reading intensity; white, fluorescent lights are recommended.

A laboratory bench and cabinet should be provided. The top should be chemically resistant and a chemically resistant sink provided with water is desirable. A distilled or deionized water source should be available. A record file for test results and references should be provided.

The equipment at each location should be selected to perform all of the tests routinely conducted. Sufficient standard laboratory equipment such as beakers, flasks and graduated cylinders should be provided. A hot plate is desirable. Analytical instruments such as pH and conductivity meters and colorimeters or spectrophotometers may be provided. All instruments must be kept clean and calibrated frequently.

Reagents must be available in sufficient, but not excessive quantities. Reagent solutions should be standardized at regular intervals and labeled with the date of each standardization. Cleanliness is a most essential property of either a field or a central laboratory.

Cooling water samples should be taken in clean glass or plastic bottles, capped immediately and tested without delay. The samples must be representative of the water in the system. The sample should not be taken immediately after treatment chemicals are added. The chemicals must be allowed to mix with the system water. Samples of the circulating water can be dipped from the cold well, or taken from the circulating pump discharge or any other convenient spot.

Section IX.
REFERENCES AND BIBLIOGRAPHY

A. CTI PUBLICATIONS:

1. Bibliography of Technical Papers, CTI Bulletin PRM-103.

   Contains a listing of all the technical papers presented at CTI meetings from 1961 to date. Over 150 water treatment related papers are included.

2. Journal of the Cooling Tower Institute

   Currently published twice each year. Contains the best papers from the CTI meetings.

3. Wood Maintenance, CTI Study WMS-104 (Rev. 1977).

4. Handling Water Treatment Chemicals Safely, CTI Procedure WTP-129 (1982).

5. Supervisory Guide - Handling Water Treatment Chemicals, CTI Procedure WTP-132 (1984).

6. Guidelines for Evaluation of Cooling Water Treatment Effectiveness, CTI Procedure WTP-130 (1982).

7. Fouling Monitors for Cooling Water Systems, CTI Technical Paper TP-85-2 (1985).

B. GENERAL REFERENCE BOOKS:

1. BETZ Handbook of Industrial Water Conditioning, Eighth Edition, Chapters 23-35, BETZ Laboratories, Trevose, PA (1980).

2. Drew Principles of Industrial Water Treatment, Fourth Edition, Chapters 3-9, Drew Chemical Corporation, Boonton, NJ (1981).

3. The Nalco Water Handbook, Chapters 20-23, McGraw-Hill, New York (1979).

4. McCoy, J.W., The Chemical Treatment of Cooling Water, Second Edition, Chemical Publishing Co., New York (1983).

5. Tanis, J.N., Procedures of Industrial Water Treatment, Chapters 23-41, Ltan, Inc. Ridgefield, CT (1987).

6. Blake, R.T., Water Treatment for HVAC and Potable Water Systems, McGraw-Hill, New York (1980).

C. PRETREATMENT:

1. Sanks, R.L., (ed.), Water Treatment Plant Design For the Practicing Engineer, Ann Arbor Science, Ann Arbor, MI (1978).

2. Hudson, H.E., Water Clarification Processes - Practical Design and Evaluation, Van Nostrand Reinhold, New York (1981).

3. Puri, V.K., Pretreatment of Water for Cooling Water and Steam Generating Systems, Calgon Corporation, Pittsburgh (1983).

4. Nordell, E., Water Treatment for Industrial and Other Uses, Reinhold, New York (1961).

5. Powell, S.T., Water Conditioning for Industry, McGraw-Hill, New York (1971).

6. AWWA, Water Quality and Treatment, McGraw-Hill, New York (1971).

7. Degremont, Water Treatment Handbook, Fifth Edition, Halsted Press (1979).

D. WATER CHEMISTRY AND ANALYSIS:

1. Faust, S.D. and Aly, O.M., Chemistry of Water Treatment, Butterworths, Boston (1983).

2. APHA, AWWA, WPCF, Standard Methods For the Examination of Water and Wastewater, Sixteenth Edition, American Public Health Association, Washington (1985).

3. ASTM, Annual Book of ASTM Standards - Section 11 - Water and Environmental Technology, American Society for Testing and Materials, Philadelphia (1987).

E. CONTROL OF MICROBIOLOGICAL ACTIVITY:

1. Cheremisinoff, N.P., P.N. and Trattner, R.B., Chemical and Nonchemical Disinfection, Ann Arbor Science, Ann Arbor, MI (1981).

2. White, G.C., Handbook of Chlorination, Second Edition, Van Nostrand Reinhold, New York (1986).

3. McCoy, J.W., The Microbiology of Cooling Water, Chemical Publishing, New York (1980).

4. Dewey, J.E., et al, Pesticide Applicator Training Manual - Category 7 - Industrial, Institutional Structural and Health - Subcategory: Cooling Towers, Cornell University, Ithaca, New York (1981).

F. CORROSION:

1. Uhlig, H.H., Corrosion and Corrosion Control, Wiley, New York (1963).
2. Speller, F.N., Corrosion: Causes and Prevention, McGraw-Hill, New York (1951).
3. Nathan, C.C., (ed.), Corrosion Inhibitors, NACE, Houston (1963).
4. NACE, Cooling Water Treatment Manual (TPC-1), National Association of Corrosion Engineers, Houston (1971).
5. MTI, State-of-the-Art Survey of Nonchromate Cooling Water Corrosion Inhibitors, Manual No. 4, Materials Technology Institute, St. Louis (1980).
6. MTI, Microbiologically Influenced Corrosion: A State-of-the-Art Review, Publication No. 13, Materials Technology Institute, St. Louis (1984).
7. Uhlig, H.H., Corrosion Handbook, John Wiley and Sons, Inc., New York (1948).

G. SCALING AND FOULING

1. EPRI, Proceedings: Condenser Biofouling Control - State-of-the-Art Symposium, EPRI CS-4339, Electric Power Research Institute, Palo Alto, CA (1985).
2. ASME, Suitor, J.W. and Pritchard, A.M., (eds.), Fouling in Heat Exchange Equipment, Publication HTD-Vol. 35, American Society of Mechanical Engineers, New York (1984).
3. Somerscales, E.F.C. and Knudsen, J.G., (eds.), Fouling of Heat Transfer Equipment, Hemisphere Publishing, Washington (1981).
4. Garey, J.F., et al (eds.), Condenser Biofouling Control, Ann Arbor, MI (1980).
5. Perry, R.H. and Green, D., Chemical Engineers' Handbook, Sixth Edition, McGraw-Hill, New York (1986).
6. EPRI, Biofouling Detection Monitor Development: Status Assessment, EPRI CS-3914, Electric Power Research Institute, Palo, CA (1985).
7. Knudsen, J.F. and Katz, D.L., Fluid Dynamics and Heat Transfer, McGraw-Hill, New York (1958).

H. BIOFOULING MONITORING

1. Kozelski, J.J., Field Experience With a Simple Cooling Water Biofilm Monitoring Device, 44th Annual International Water Conference, Paper 46, October 1983.
2. Strauss, S.D., Biofouling: New Insights, New Weapon, Power, April 1981.
3. Zuniga, P., Miller, K., Winters, M., Search for a Better Cooling Water Monitor, 50th Annual International Water Conference, Paper 34, October 1989.
4. McCoy, W.F., Bryers, J.D., Robbins, J., Costerton, J.W., Observations of Fouling Biofilm Formation, Can. J. Microbiology, Vol. 27, 1981.

Section X.
GLOSSARY OF TERMS

1. Acidity: A quantitative capacity of aqueous media to react with hydroxl ions to form water, expressed as ppm or mg/l.
2. Adsorption: Assimilation of molecules or other substances into the physical structure of a liquid or solid without chemical reaction.
3. Aeration: Blowing air through water to sweep out other dissolved gases and to equilibrate the water with oxygen and carbon dioxide.
4. Aerobic: An organism that requires oxygen for its respiration.
5. Aerosol: A colloidal system involving liquid or solid particulates dispersed in air.
6. Agglomerate: To gather fine particulates together into a larger mass.
7. Algae: Simple plants containing chlorophyll. Many are microscopic, but under favorable conditions they can grow in colonies and produce mats and similar nuisance masses.
8. Alkalinity: An expression of the total basic anions (hydroxyl groups) present in a solution. It also represents, particularly in water analysis, the bicarbonate, carbonate, and occasionally, the borate, silicate, and phosphate salts which will react with water to produce hydroxyl groups.
9. Amoeba: A microscopic one-celled animal consisting of a naked mass of protoplasm constantly changing in shape as it moves and engulfs food.
10. Anaerobic: An organism that can thrive in the absence of oxygen.
11. Anion: A negatively charged ion.
12. Anodic Area: Area on a metal surface where electrons are given up and metal dissolves (corrosion).
13. Antinucleation: Process by which the formation of a nuclei is prevented or inhibited.
14. Atomic Weight: The weight of an atom of any element compared with that of oxygen which is considered to be 16.
15. Bacteria: Microscopic single-cell plants which reproduce by fission or by spores, identified by their shapes: coccus, spherical; bacillus, rod-shaped; and spirillum, curved.
16. Backwash: That part of the operating cycle of an ion-exchange process in which a reverse upward flow of water expands the bed, effecting such physical changes as loosening the bed to counteract compacting, stirring up and washing off light insoluble contaminants to clean the bed, or separating a mixed bed into its components to prepare it for regeneration.
17. Base: An alkaline substance.
18. Bicarbonate Alkalinity: The presence in a solution of hydroxyl (OH^-) ions resulting from the hydrolysis of carbonates or bicarbonates. When these salts react with water, a strong base and a weak acid are produced, and the solution is alkaline.
19. Biochemical Oxygen Demand (BOD): The quantity of oxygen required for the biological and chemical oxidation of water-borne substances under conditions of test.
20. Biocide: A chemical used to control the population of troublesome microbes.
21. Biological Deposits: Water formed deposits of organisms or the products of their life processes.

    Note: The biological deposits may be composed of microscopic organisms, as in slime, or of macroscopic types such as barnacles or mussels. Slimes are usually composed of deposits of a gelatinous or filamentous nature.
22. Bleedoff: (See Blowdown)
23. Blowdown: The withdrawal of water from an evaporating water system to maintain a solids balance within specified limits of concentration of those solids.
24. Brackish Water: Water having a dissolved matter content between that commonly observed for fresh water and that of sea water.
25. Brine: Water having more than approximately 30,000 mg/liter of dissolved matter.
26. BOD: (See Biochemical Oxygen Demand)
27. Cation: A positively charged ion resulting from dissociation of molecules in solution.
28. Carbonate Hardness: That hardness in water caused by bicarbonates of calcium and magnesium. If alkalinity exceeds total hardness, all hardness is carbonate hardness; if hardness exceeds alkalinity, the carbonate hardness equals the alkalinity.
29. Cathodic Area: Area where electrons are accepted and cathodic reactions occur, i.e.,

reduction of dissolved oxygen to form hydroxyl ions.

30. Chelation: Solubilization of scale forming materials by a stoichimetric reaction that locks the metal ion into an organic ring structure. Example, EDTA.

31. Chemical Oxygen Demand (COD): A measure of organic matter and other reducing substances in water.

32. Chlorine Demand: The total amount of chlorine required to just achieve a residual in a given system.

33. Chlorine, Free: (Sometimes called residual chlorine) The amount of chlorine remaining in water after reaction with nitrogen or nitrogenous compounds

34. Chlorine, Combined: Residual consisting of chlorine combined with ammonia nitrogen or nitrogenous compounds.

35. Chlorine Total: The amount of chlorine required to react with ammonia nitrogen or nitrogenous compounds plus the free chlorine residual.

36. Chlorination: The addition of chlorine or a chlorine-releasing agent to water.

37. Coagulation: The neutralization of charges on colloidal matter to allow particles to grow and settle.

38. COD: (See Chemical Oxygen Demand)

39. Colloids: Matter of very fine particle size, usually in the range of 10^{-5} to 10^{-7} cm in diameter.

40. Concentration: The process of increasing the dissolved solids per unit volume of solution, usually by evaporation of the liquid; also, the amount of material dissolved in a unit volume of solution.

41. Concentration Cell: The connection of two solutions of the same composition but different concentrations by a metal conductor to produce current flow through the circuit.

42. Concentration Ratio: In an evaporating water system, the ratio of the concentration of a specific substance in the circulating water to its concentration in the makeup water.

43. Concentration, Cycles of: Concentration ratio.

44. Conduction: The transfer of heat through a body by molecular motion.

45. Conductivity: The ability of water to conduct electricity. When measured with a standard apparatus, it is called specific conductivity and is a function of the total dissolved solids. As a rule of thumb, TDS = 2/3 specific conductance in micromhos (microsemens).

46. Corrosion Products: Deposits or solids formed as a result of a chemical or electrochemical reaction between a metal and its environment.

47. Crystal Distortion: Any change in the normal crystal form of a precipitate.

48. Cycles of Concentration: (See Concentration Ratio)

49. Deflocculation: The ability of some materials, such as polyphosphates, to peptize, deflocculate, and disperse suspensions of colloidal particles. See dispersants.

50. Delignification: The dissolving of the lignin portion of cooling tower wood, usually by alkaline agents.

51. Diatoms: Organisms related to algae, having a brown pigmentation and a siliceous skeleton.

52. Dispersant: A chemical which causes particulates in a water system to remain in suspension.

53. Disinfection: Application of energy or chemicals to kill pathogenic organisms.

54. Drift: Entrained water in the stack discharge of a cooling tower.

55. Dissolved Solids: (See Total Dissolved Solids)

56. Effluent: Final discharge from any process.

57. Electrolyte: A chemical compound which dissociates or ionizes in water to produce a solution which will conduct an electric current; an acid, base, or salt.

58. Entrainment: (See Drift)

59. EPM: Equivalents per million; parts per million of any ion divided by its equivalent weight.

60. Equivalent Weight: Molecular or atomic weight divided by valence.

61. Excursion: Any change in cooling water characteristics that takes it outside of control limits.

62. Flocculation: The process of agglomerating coagulated particles into settleable flocs, usually of a gelatinous nature.

63. Fluidizers: Chemicals that distort crystal structures so that the crystals do not stick to surfaces or to each other.

64. Fouling: The deposition on heat transfer surfaces, or other surfaces, of material normally dissolved or suspended in cooling water.

65. Free Mineral Acidity: The quantitative capacity of a water to react with hydroxyl ions up to pH 4.3.

66. Fungi: As applied to water, simple, one-celled organisms without chlorophyll, often filamentous. Molds and yeasts are included in this category.

67. Grains Per Gallon: An expression of concentration of material in solution, generally in terms of calcium carbonate. One grain (as calcium carbonate) per gallon is equivalent to 17.1 parts per million.

68. Ion: Any dissolved particle in solution possessing either a positive or a negative electric charge.

69. Hardness, Calcium: The calcium compounds dissolved in water, expressed as calcium carbonate.

70. Hardness, Carbonate: The calcium and magnesium carbonate and bicarbonate dissolved in water, expressed as calcium carbonate.

71. Hardness, Magnesium: Magnesium compounds dissolved in water, expressed as calcium carbonate.

72. Hardness, Non-carbonate: The difference between the total hardness and the total alkalinity of a water.

73. Hardness, Permanent: The hardness that cannot be removed from water by boiling. Essentially, the same thing as the non-carbonate hardness.

74. Hardness, Temporary: The hardness that can be removed from water by boiling. Essentially, the same as the carbonate hardness.

75. Hardness, Total: The sum of the calcium and the magnesium hardness. Also the sum of the permanent hardness and temporary hardness. The U.S. Geological Survey hardness criteria for potable water are: Softwater 0-60 ppm, moderately hardwater 61-120 ppm, hardwater 121-180 ppm, very hard water >180 ppm.

76. Heavy Metal: An arbitrary name given to certain metallic ions which are toxic to aquatic organisms. They include chromium, zinc, mercury and lead.

77. Hydroxyl: (See Alkalinity)

78. Langelier's Index: A formula for predicting whether water will tend to dissolve or precipitate calcium carbonate. If the water precipitates calcium carbonate, scale formation may result. To calculate Langelier's Index, the actual pH value of the water and Langelier's saturation pH value (pH_S) are needed. pH_S is determined by the relationship between the calcium hardness, the total alkalinity, the total solids concentration and the temperature of the water. Langelier's Index is then determined from the expression pH minus pH_S. If pH minus pH_S is positive, calcium carbonate may precipitate in the system. If pH minus pH_S is negative, calcium carbonate may dissolve.

79. mg/l: Milligrams per liter; essentially, the same as parts per million when applied to water solutions.

80. Molecular Weight: The sum of the atomic weights of the atoms in a molecule.

81. Nonionic: A colloidal particle without a surface charge, or an unionized molecule in solution.

82. Non-Oxidizing Biocide: A biocide whose effectiveness depends upon some property other than its ability to oxidize organic material, i.e., systemic poisons and surface activity.

83. Oxidizing Biocide: A biocide whose effectiveness depends upon its ability to oxidize, and thus destroy organic material. Example: chlorine, bromine, ozone.

84. ORP: Oxidization-reduction also called the Redox potential. ORP is the potential of an electrode in water relative to a standard hydrogen or calomel electrode. ORP can be used to measure the concentration of an oxidizing biocide in water, or to indicate the corrosivity of the water.

85. Particulate: Any suspended, or undissolved particle in water.

86. pH: The logarithm of the reciprocal of the hydrogen ion concentration in water. It is a measure of acidity or alkalinity. Water with a pH of 7.0 is neutral at room temperature. A pH greater than 7.0 indicates an alkaline water.

87. (pH): Equilibrium pH; the pH a circulating water assumes when in equilibrium with the contacting air. It is a function of alkalinity and temperature.

88. $(pH)_S$: pH of saturation; the pH above which a material will precipitate. In dealing with cooling water, it is assumed to be the pH of saturation of calcium carbonate. This term is used in the calculation of the Langelier and Ryznar indices.

89. PPM: Parts per million; a part per million is one pound (or any other weight unit) per million pounds (or other weight unit) of water.

90. Polyelectrolyte: Any water-soluble polymer used for clarification, either by coagulation or flocculation.

91. Precipitation: Formation of a solid material when a compound dissolved in water exceeds its solubility limit.

92. Phosphates: Inorganic ions comprised of phosphorus and oxygen.

93. Quaternary Ammonium: A specific basic group [$-N(CH_3)_4^+$] on which depends the exchange activity of certain strong base anion exchange resins. Some biocides contain quaternary ammonium salts.

94. Recarbonation: The injection of carbon dioxide into water, usually after lime softening, to decrease the pH of the water.

95. Ryznar Index: An empirical method for predicting scaling tendencies of water based on a study of operating results with water of various saturation indices.

Stability Index = $2pH_S$-pH

where:

pH_S = Langelier's Saturation pH

This index is often used in combination with the Langelier Index in predicting the scaling or corrosion tendencies of a water. If the stability index is between 5.5 and 6.5, the water is said to be neutral, or stable. Lower values indicate that calcium carbonate will precipitate, and higher values indicate that calcium carbonate will dissolve.

96. Saturation Index: The same as the Langelier Stability Index (LSI).

97. Scale: The deposition on heat transfer surfaces, of material normally in solution, as opposed to fouling, which is deposition of material normally in suspension.

98. Sedimentation: Gravitational settling of solid particles in a liquid system.

99. Sequestration: The formation of water soluble complexes with deposit-forming ions. Chelation is one form of sequestration.

100. Softening: Reduction of hardness.

101. Solubilization: Anything, including pH reduction and sequestration, that increases the solubility of deposit-forming materials.

102. Specific Conductivity: (See Conductivity)

103. Stability Index: The same as the Ryznar Index.

104. Stoichiometric: The ratio of chemical substances reacting in water that corresponds to their combining weights in the theoretical chemical reaction.

105. Surfactant: In water, any molecule that modifies the interfacial tension of a liquid and the surface it contacts (surfaces such as air, metal, biomass, etc.)

106. Suspended Solids: Insoluble particles suspended in water.

107. Synergism: The condition existing where the effect of two or more materials added to water is greater than the sum of their individual effects.

108. TDS: Total dissolved solids; the sum of the inorganic and organic materials dissolved in water.

109. Threshold Concentration The critical concentration of an additive that must be maintained in order to control/prevent a given reaction (such as the prevention of calcium carbonate scale formation on heat transfer surfaces).

110. Total Organic Carbon (TOC): The carbon contained in organic materials dissolved in water, but not including carbonates, bicarbonates, or carbon dioxide.

111. TOC: (See Total Organic Carbon)

112. Tuberculation: Localized attack typified by the formation of inverted pliable conical structures. In water systems, tuberculation is most commonly associated with localized corrosion of iron and the conical structures are predominantly made up of several oxidation states of iron (Fe_2O_3, Fe_3O_4, etc.)

113. Turbidity: The interference to light transmission caused by suspended solids, usually colloidal in nature.

114. Valence: The number of charges, either positive or negative, associated with an ion.

115. Water Formed Deposit: Any accumulation of insoluble material derived from water or formed by the reaction of water upon surfaces, including scale, sludge, foulants, sediments, corrosion products, or biological deposits.

COOLING TECHNOLOGY INSTITUTE

PO Box 73383 Houston, Texas 77273
281.583.4087   Fax 281.537.1721
www.cti.org   email: vmanser@cti.org